Reviewing Backfilled SOFR

Ram Prasad

Introduction 2
Models 3
Data 6
Evidence of Deficiencies in Backfilled SOFR 8
Revised Testing Framework 13
Model Estimation and Performance 15
Arithmetic Brownian Motion (MDD) 15
White Noise Model 18
Average Spread Model 19
Zero Spread Model 20
PCA Model 21
Alternative Data Source 24
Discussion 27
Conclusion 29
Appendix 30
Fat-Tail Scaling 30
Zero Backfilled SOFR-EFFR Spreads 32
SOFR-EFFR Spread Summary Statistics 33
Observed SOFR-EFFR Spread Distributions 34
SOFR-EFFR Spread Cross-Tenor Correlation 35
Correlation Matrix Difference SOFR EFFR Historical and Backfilled 36
SOFR Training, Testing, Validation Period Summary Statistics 37
Time-Series Properties of SOFR-EFFR Spread 38
ABM Random Draw Variation 42
Standardized PCR Results 47
Full Sample Regression Test Results 48

Introduction

LIBOR, the London Interbank Offered Rate, was a survey-based benchmark short-term reference rate. It underpinned transactions with $200 trillion in notional, despite less than $500 million in underlying daily transactions, and was replaced owing to manipulation scandals. SOFR, the secured overnight financing rate, is a market-based rate that measures observed overnight borrowing costs in the US Treasury repo market. Specifically, SOFR is a volume-weighted median of repo rates observed in a single business day, drawing on tri-party, FICC GCF, and bilateral Treasury repo transaction data, and reported the following morning by the Federal Reserve Bank of New York. SOFR's publication began on April 3rd, 2018.
uses historical time series of reference rates to calibrate covariance matrices for the multivariate distribution of market factor returns. Stress periods prior to 2018, such as the 2007-2009 period, are consequently of key interest; to evaluate the sensitivity of interest rate derivatives currently in portfolios to market conditions in key stress periods, developed a methodology to backfill SOFR and other alternative reference rates. These imputed time series are fed into volatility and correlation calculations and in turn bear heavily on SVaR.

The goal of this exercise is to assess the existing methodology for backfilled SOFR, as described in the model development document (subsequently, MDD),
(2023), outline its merits, deficits, directions for improvement, and produce a superior backfilled SOFR series. The MDD is an exhaustive treatment of multiple alternative reference rates, going well beyond the consideration of SOFR. For brevity, I consider the issue of backfilled SOFR itself, but my results and conclusions bear on the exercise of backfilling other rates. This paper proceeds as follows. First, I expand on the mathematical foundations of the current proxy plus backfilled spread methodology and document the issue of increasing spread ensemble variance. After performing quality checks on the reference rate source data, I confirm the suitability of the effective federal funds rate as a proxy for SOFR and identify several deficiencies in the current backfilled SOFR series. A key concern for any backfilled series is what constitutes a suitable testing framework. I argue for partitioning the post-April 2018 period in which the SOFR index was published and observed into backwards looking temporal testing and training periods to directly assess model performance. This approach is a stark departure from the MDD's current set of assessments and demonstrates that the magnitude of errors in spreads distorts the backfilled SOFR series. I use several models to backfill SOFR by predicting backwards over the 2007 - 2018 period, including models for the SOFR - EFFR spread per tenor (arithmetic Brownian motion, white noise, a constant average spread, and zero spread) and for SOFR itself (principal components regression). I report error measures for the spread where applicable, and for SOFR itself, per tenor. After considering each models deficits, I recommend an alternative time series cited by the Federal Reserve Bank of New York, which is a combination of a FRBNY survey rate and the DTCC GCF rate, as a direct proxy for 1-month SOFR. Finally, I discuss issues, some inherent to any backfilling exercise and some specific to the LIBOR transition at that constrain the quality of backfilled SOFR.

Ultimately, I conclude that the current approach should not be used and recommend using the FRBNY model for backfilling 1-month SOFR, average spreads for short tenor SOFR, and principal components regression for long-tenor SOFR.

Models

I outline the current model for backfilled SOFR, as stated in the MDD, and add some detail. Let time t_0 be the first point at which SOFR is observed[1]. For t_0 and all subsequent time periods, the SOFR time series is observed historical SOFR. Prior to t_0, backfilled SOFR per tenor equals a proxy measure. The proxy takes the form of the same tenor effective federal funds rate plus a stochastically modelled SOFR - EFFR spread. The τ superscript denotes tenor, such as 1 month, 12 months, and 360 months.

$$\widehat{\mathrm{SOFR}}_t^{\tau} = \mathbf{1}_{\{t \geq t_0\}} \cdot \mathrm{SOFR}_t^{\tau} + (1 - \mathbf{1}_{\{t \geq t_0\}}) \cdot \mathrm{Proxy}_t^{\tau}$$

$$\mathrm{Proxy}_t^{\tau} = \mathrm{EFFR}_t^{\tau} + S_t^{\tau}$$

A time-varying spread has no inherent restrictions. It can be positive, negative, zero, fluctuate with no relation to the level of EFFR, or exhibit minimal variance. As a separate model, it may also be assumed to be constant. The MDD assumes the spread follows a driftless arithmetic Brownian motion[2]. Subsequent notation includes a constant drift term for clarity, as the imposed assumption of zero drift may or may not be empirically reasonably going forward. With per-tenor parameters, the spread has the dynamics:

$$dS_t^{\tau} = \alpha^{\tau} dt + \sigma^{\tau} dW_t^{\tau}$$

$$S^{\tau}(t) - S^{\tau}(0) = \int_0^t \alpha^{\tau} ds + \int_0^t \sigma^{\tau} dW^{\tau}(s)$$

$$S_t^{\tau} = S_0^{\tau} + \alpha^{\tau} t + \sigma^{\tau} W_t^{\tau}$$

$$S_t^{\tau} \sim N(S_0^{\tau} + \alpha^{\tau} t, \sigma^{\tau 2} t)$$

where the initial spread S_0^{τ} is the spread at t_0 and $W_t^{\tau} \sim N(0,t)$. Each tenor has its own constant drift, diffusion term and Brownian motion[3]. The latter permits modelling correlation between spread changes across tenors. For an assumed drift of $\alpha=0$, the expected spread at any time is the initial spread and the variance of the spread, $\sigma^{\tau 2}t$, increases linearly with time. This presents two unavoidable problems. First, the spread on average equals an initial spread observed just once, instead of a smoother function of subsequent observed spreads. Second, the spread is more volatile spread the farther SOFR is backfilled. Repeating the backfilling process should result in spreads with increasing ensemble variance. The one period spread change establishes a recursive relationship to backfill spreads by drawing unit normals ε_t^{τ}

$$\begin{aligned} S_{t+1}^{\tau} - S_t^{\tau} &= S_0^{\tau} + \alpha^{\tau}(t+1) + \sigma^{\tau} W_{t+1}^{\tau} - (S_0^{\tau} + \alpha^{\tau} t + \sigma^{\tau} W_t^{\tau}) \\ &= \alpha^{\tau} + \sigma^{\tau}(W_{t+1}^{\tau} - W_t^{\tau}) \\ S_t^{\tau} &= S_{t+1}^{\tau} - \alpha^{\tau} - \sigma^{\tau}(W_{t+1}^{\tau} - W_t^{\tau}) \end{aligned}$$

and subtracting their sums. S_t^{τ} for $t < t_0$ denotes the backfilled spread. With drift, it equals:

$$\begin{aligned} S_{t<t_0}^{\tau} &= S_0^{\tau} - t\alpha^{\tau} - \sigma^{\tau} \sum_{s=t+1}^{t_0} c_s^{\tau} \\ &= \mathrm{SOFR}_{t_0}^{\tau} - \mathrm{EFFR}_{t_0}^{\tau} - t\alpha^{\tau} - \sigma^{\tau} \sum_{s=t+1}^{t_0} c_s^{\tau} \end{aligned}$$

A Brownian motion is stationary with independent increments. Consequently, backfilled spreads per tenor should exhibit zero serial correlation and may exhibit cross-tenor correlation. Modelling time

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[1] SOFR is first available from April 3rd, 2018. The MDD states t_0 is Sept. 4th, 2018 using [illegible] as the data source. I have found multiple references to FRBNY pre-production estimates of SOFR, spanning Aug. 2014 – Mar. 2018, but couldn't obtain the actual pre-production series.

[2] A geometric Brownian motion models non-negative spreads. Empirically, the spread between SOFR and EFFR can be negative (SOFR-EFFR < 0).

[3] In the MDD, there is no tenor superscript for the diffusion of Brownian motion but the description of correlated spread changes across tenors and volatility of absolute spread changes requires tenor specific Brownian motions.

varying cross-tenor correlation can be done as follows[4]. For arbitrary tenors τ_1 and τ_2, their Brownian motions satisfy the relation $dW_t^1\, dW_t^2 = \rho^{12}_t dt$ such that the change in spreads have instantaneous correlation of $\rho^{12}_t\, \sigma^1_t\, \sigma^2_t\, dt$. Time-varying instantaneous correlation, as shown below, is not the stated

$$\begin{aligned} dS_t^1 &= \alpha^1 dt + \sigma^1 dW_t^1 \\ dS_t^2 &= \alpha^2 dt + \sigma^2 dW_t^2 \\ dS_t^1\, dS_t^2 &= \sigma^1 dW_t^1 \sigma^2 dW_t^2 \\ &= \sigma^1 \sigma^2 \rho_t^{1,2}\, dt \end{aligned}$$

formulation in the MDD. The MDD uses draws from a multivariate normal distribution, where the covariance matrix is constant and calibrated using observed cross-tenor correlation. This implies a much simpler model, which is to assume that the cross-section of spreads has a multivariate normal distribution with possibly time-varying moments. The variance of the spread now does not increase with

$$\begin{aligned} \mathbf{S_t} &\sim N(\mu_t, \Sigma_t) \\ S^\tau_{t<t_0} &= \mu^\tau + \sigma^\tau \epsilon^\tau_t \end{aligned}$$

time alone, and the simulation procedure is akin to that of the ABM process but without requiring sequentially backfilling the spread level from the initial observed spread at t_0. A single aberrational spread draw no longer impacts all subsequent backfilled spreads. I implement this model with a time-invariant mean and covariance matrix and use the method of moments estimators (the sample mean and covariance matrices, respectively)[5]. Backfilled spreads should exhibit no serial correlation.

I experiment with constant spread models as well, using the average observed spread and a zero spread. In the average model, the constant spread pre tenor equals the simple arithmetic average of observed SOFR – EFFR per tenor. In the zero-spread model, the spread at all tenors is 0, so backfilled SOFR per tenor is EFFR at the corresponding tenor. Note the lack of a spread time subscript below.

$$\text{Proxy}^\tau_t = \text{EFFR}^\tau_t + S^\tau$$

Finally, I replicate the principal components regression methodology of the MDD. Assume that SOFR is a function, possibly non-linear, of observable macroeconomic time series, such as EFFR at various tenors. As used in empirical asset pricing, PCA purports to uncover a cross-tenor factor structure in bond yields by identifying a few linear combinations of yields that explain most of the variation in yields. The first three principal components are referred to as the level, slope, and curvature factors, respectively. PCR models SOFR itself, not the SOFR - EFFR spread, as a linear combination of these factors, and estimates the coefficients by regressing observed SOFR per tenor on the principal component scores. Backfilled SOFR equals a linear combination of regression-estimated betas and principal component scores, where pre-t_0 scores are computed using the factor loadings obtained from PCA on post-t_0 EFFR. M, the number of principal components, is a hyperparameter to be selected using some form of cross-validation.

$$\text{SOFR}^\tau_t = \sum_{k=1}^{M} \beta^\tau_k \phi_{k,t} + \epsilon_t$$

$$\widehat{\text{SOFR}}^\tau_{t<t_0} = \sum_{k=1}^{M} \hat{\beta}_k^{\tau,t\geq t_0} \phi_{k,t<t_0}$$

[4] MDD documentation is sparse on the matter of cross-tenor spread correlation. It describes draws from a multivariate normal with an correlation matrix of spread change innovations. A zero-spread model preserves the cross-tenor correlation of EFFR.

[5] Effectively, the spread is white noise. A broader question is why the MDD uses a stochastic process like an arithmetic Brownian motion instead of a time series model for the spread. The spread at each tenor can then be modelled as an ARIMA(p,d,q) model, with lags and difference order estimated conventionally. For example, white noise is an ARIMA(0,0,0) model. The choice of model should reflect economic intuition about the spread between two reference rates; in the absence of such priors, I evaluate each model and use test error measures to discriminate.

Unlike the spread models, the PCA model produces an underlying structure for the benchmark rate from a range of EFFR tenors. The spread models are EFFR at the same tenor plus a spread, with no meaningful economic interpretation[6] for the spread. A durable factor structure can be quite useful for backfilling.

Summary of Models Evaluated

Model	Description
$SOFR^{\tau}_{t<t0} = EFFR^{\tau}_{t} + S_t^{\tau}$	
Arithmetic Brownian Motion	Spreads are sequentially backfilled per tenor by subtracting scaled normal draws from the initial spread.
White Noise	Spreads equal draws from a multivariate normal distribution, with a covariance structure calibrated to 9/4/18 – 12/31/20 data.
Average	Spreads equal the 9/4/18 – 12/31/20 average spread per tenor.
$SOFR^{\tau}_{t<t0} = f(EFFR^{1}_{t} \ldots EFFR^{600}_{t})$	
PCA	Backfilled SOFR is a linear combination of estimated betas and principal component scores, where betas and loadings are estimated from 9/4/18 – 12/31/20 data.
Zero	All spreads are 0, so backfilled SOFR is historical EFFR.

Models for backfilled SOFR. Spread models estimate S_t^{τ}, the difference between SOFR and EFFR per tenor. SOFR models backfill SOFR directly. The zero model is a baseline. It is equivalent to setting a constant spread of 0 per tenor.

Modelling SOFR directly as a function of observed time series is a regression problem, for which there are many more models than I examine here. I evaluate PCA because of its prevalence in the interest rate term structure literature and use the zero-spread model as a baseline. Although it is possible to use other proxies for SOFR instead of EFFR, such as LIBOR, I use EFFR as it exhibits both strong co-movement with and similarity in levels to SOFR. Further, it is cited as a proxy for SOFR in the MDD as well as in external literature[7]. I limit the set of covariates to reference rates for brevity.

[6] Level, slope, and curvature refer to the shape of principal components, and can be observed by plotting the eigenvectors (loadings against tenors). These commonly observed shapes may express a foundational component of the term structure or may be a statistical artifact of PCA. Even if this structure is merely a product of highly correlated term structures, bereft of economic meaning, it merits consideration.

[7] See D. Bowman, "Historical Proxies for the Secured Overnight Financing Rate" (2019). FEDS Notes, Federal Reserve Board of Governors.

Data

retains historical time series for reference rates, in a database called requested time series of 1-month to 480-month LIBOR[8], 1-month to 600-month Effective Federal Funds Rate, and 1-month to 600-month Secured Overnight Financing Rate. The SOFR series include both historical and backfilled values, but the raw data does not separate the two. It isn't obvious where the series changes from observed historical to backfilled values, and whether that delineation is consistent

Reference Rates

Series	Range	T	Tenors
LIBOR	1/2/07 – 12/31/20	3411	1M – 480M
SOFR	1/3/07 – 12/31/20	3521	1M – 600M
EFFR	1/3/07 – 8/31/23	4203	1M – 600M

Reference rate data. Source:

across tenors. All data is in percentage points, so a recorded value of 5.32 means 5.32% and a spread of 0.04, for example, means 0.04%. I check each series for data quality, including negative rates, staleness, and outliers[9]. Daily SOFR values can be negative as FRBNY does not have a floor for the published SOFR index. The smallest negative SOFR value for the 24-month tenor is -0.159%, which occurs on 8/10/11. For the 2M, 3M, 6M, 9M, and 12M tenors, negative SOFR rates appear in the pre-2018 backfilled period. For the 18M, 24M, 36M, and 48M tenors, negative SOFR appear as late as Aug. 2020. Short-tenor SOFR was near zero per the FRBNY website in early Aug. 2020, so these may not be data quality issues. Negative EFFR values are more puzzling, as the nominal effective federal funds rate is a volume weighted median of overnight transactions, and hence non-negative, unless the Fed charges depository institutions for their reserve balances. The EFFR series appear to be nominal, suggesting minor data quality issues for the 1M – 8M, 24M, and 36M tenors. Staleness is noticeable for EFFR and LIBOR, but this is to be expected, as EFFR is tied to the target federal funds rate, which resembles a step function, and LIBOR is a survey rate. SOFR, a market-based rate, naturally exhibits less staleness. There is 1 outlier value for the 1M and 2M tenors in EFFR, using the criteria of 8 standard deviations from the mean.

Reference Rate Data Quality Checks

Tenor	Count of Negative Rates			Staleness Count			Outlier Count		
	SOFR	EFFR	LIBOR	SOFR	EFFR	LIBOR	SOFR	EFFR	LIBOR
001M	0	68	0	50	440	1138	0	1	0
002M	19	25	0	1	58	416	0	1	0
003M	126	23	0	4	293	938	0	0	0
006M	19	14	0	2	17	252	0	0	0
009M	26	8	0	2	8	168	0	0	0
012M	15	0	0	1	5	126	0	0	0
018M	40	1	0	1	4	83	0	0	0
024M	180	13	0	0	3	62	0	0	0
036M	27	10	0	0	0	50	0	0	0
048M	3	1	0	1	3	34	0	0	0
060M	0	0	0	0	11	27	0	0	0
084M	0	0	0.0	0.0	13	40	0	0	0

[8] LIBOR has 15M and 21M tenors as well. SOFR and EFFR have the same tenors.

[9] Staleness is the count of observations for which the absolute 1-day change in levels is less than 10^{-6}. Outlier is the count of observations for which their absolute z-scores exceed 8, where the mean and standard deviation are computed using a rolling 1-year (T=252) window.

120M	0	0	0.0	2.0	13	26	0	0	0
180M	0	0	0.0	2.0	12	21	0	0	0
240M	0	0	0.0	4.0	14	24	0	0	0
300M	0	0	0.0	3.0	15	24	0	0	0
360M	0	0	0.0	3.0	19	29	0	0	0
480M	0.0	0.0	0.0	3.0	20	25	0	0	0
600M	0.0	0.0	-	1.0	18	-	0	0	-

Count of negative, stale, and outlier values in SOFR, EFFR, and LIBOR time series, per tenor. LIBOR spans 1/2/7-12/31/20 (T=3411), SOFR, 1/3/7-12/31/20 (T=3521), EFFR, 1/3/7-8/31/23 (T=4203). Outlier tests use 251 fewer observations. Data from

I spot check values from each time series against readily available data sources, such as FRED and FRBNY, and find that the series values are for the most part close but sometimes not an exact match. There are no glaring data quality issues in the reference rates' observed historical time series. Ultimately, my goal is to replicate and improve the existing methodology, so for consistency I proceed with the data provided by

Evidence of Deficiencies in Backfilled SOFR

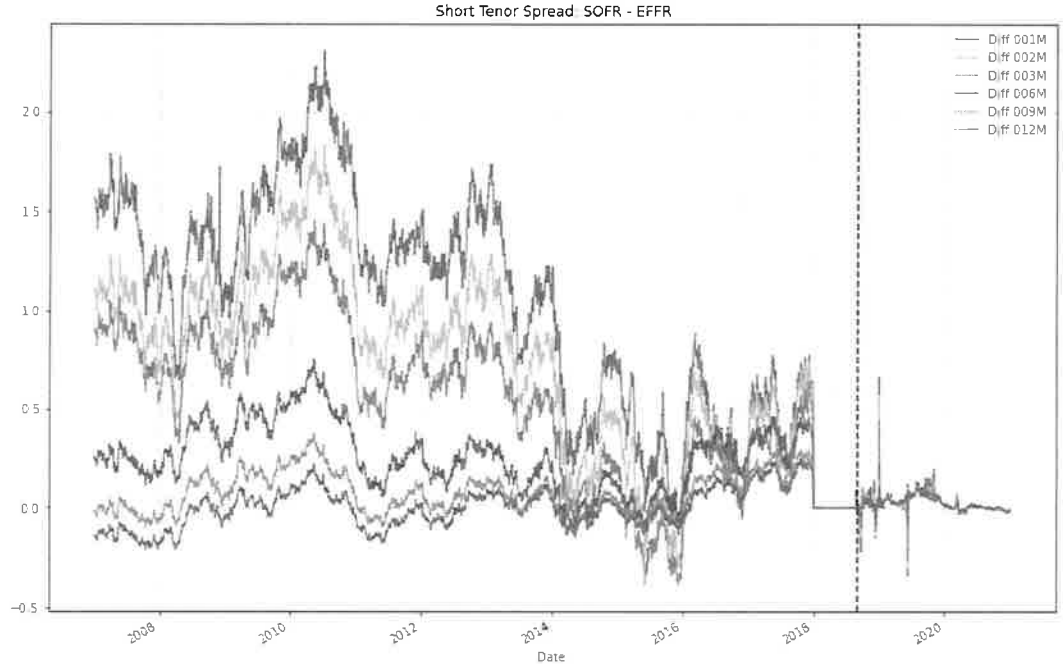
A unique feature of the assumed arithmetic Brownian motion process for the dynamic SOFR – EFFR spread is that the spread's variance increases over time. The spread's variance, $\sigma_\tau^2 t$ increases linearly with time and is larger for more volatile tenor spreads[10]. I compare summary statistics for the spread backfilled over 1/03/2007 – 08/31/2018 to the historical spread from 9/04/2018 – 12/31/2020. For every tenor, the backfilled variance is considerably higher than actual variance, up to 8 times as much.

Spread Volatility, Backfilled vs. Historical

Tenor	Backfilled Std.	Historical Std.	Ratio	Tenor	Backfilled Std.	Historical Std.	Ratio
1M	0.582	0.067	8.68	48M	0.104	0.018	5.78
2M	0.472	0.056	8.43	60M	0.111	0.018	6.17
3M	0.406	0.047	8.64	84M	0.097	0.019	5.11
6M	0.177	0.035	5.06	120M	0.085	0.022	3.86
9M	0.106	0.031	3.42	180M	0.055	0.025	2.20
12M	0.098	0.028	3.50	240M	0.060	0.027	2.22
18M	0.088	0.024	3.67	300M	0.066	0.029	2.28
24M	0.114	0.022	5.18	360M	0.071	0.031	2.29
36M	0.090	0.019	4.74	480M	0.074	0.033	2.24

Sample standard deviation of SOFR – EFFR spread per tenor across backfilled period (T=2935, 1/3/7-8/31/18) and observed historical period (T=577, 9/4/18-12/31/20). Ratio equals backfilled standard deviation divided by historical standard deviation.

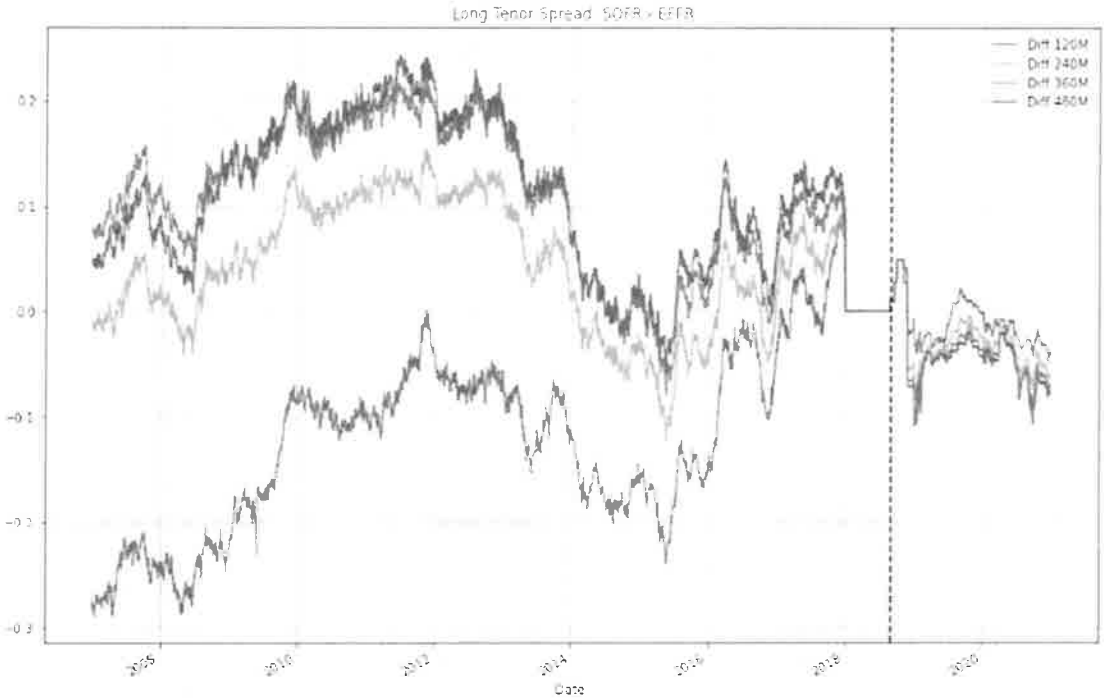
Short tenor average spreads in the backfilled period are much higher than observed historical spreads[11], and backfilled long tenor spreads are positive compared to negative observed historical spreads.



1-,2-,3-,6-,9-,12-month SOFR-EFFR spreads, T=2935, 1/3/7-8/31/18. Vertical line at 9/4/18 delineates backfilled from observed.

[10] The variance of a spread that follows a Geometric Brownian motion is $S_0^2\exp(2\mu t)[\exp(\sigma^2 t) - 1]$ and is increasing in t at an increasing rate.

[11] See the Appendix for complete summary statistics.



120-,240-,360-,480-month SOFR-EFFR spreads, T=2935, 1/3/7-8/31/18. Vertical line at delineates backfilled from observed.

Time series plots of the spreads also show a curious phenomenon. Spreads break sharply from non-zero at the beginning of 2018 to zero and then back to non-zero around September, for all tenors. I compute summary statistics for the backfilled SOFR – EFFR spread per tenor between 1/1/18 and 8/27/18, inclusive. At 3 decimal places, every statistic – average, standard deviation, maximum, and all percentiles – register at 0.000. Curiously, spreads exhibit a structural break on two dates: 12/29/17 - 1/2/18 and 8/27/18 - 8/28/18[12]. This anomaly cannot be explained by negligible spreads in the first few iterations of the arithmetic Brownian motion process, because the spreads go from non-zero to zero to non-zero. The graphs also show that backfilled SOFR – EFFR spreads do preserve the high cross-tenor spread correlation evident in the observed period, which is encouraging.

As done in the MDD, I compute the difference between the correlation of SOFR and proxy rates – EFFR and LIBOR – at corresponding tenors, in the observed period of 9/4/18 – 12/31/2020 and the backfilled period of 1/3/2007 – 9/4/18 (exclusive): $\rho^{\text{SOFR,EFFR}}{}_{\tau,t\geq t0} - \rho^{\text{SOFR,EFFR}}{}_{\tau,t<t0}$ and $\rho^{\text{SOFR,LIBOR}}{}_{\tau,t\geq t0} - \rho^{\text{SOFR,LIBOR}}{}_{\tau,t<t0}$, and as a baseline for comparison , $\rho^{\text{EFFR,LIBOR}}{}_{\tau,t\geq t0} - \rho^{\text{EFFR,LIBOR}}{}_{\tau,t<t0}$. Correlation is a measure of co-movement, and a positive difference indicates correlation between observed SOFR and a proxy rate exceeds the correlation between backfilled SOFR and the proxy rate. A large difference in magnitude indicates poor backfilling or a structural break in the relationship. There is no decisive reason to suggest the latter, indicated by the meager EFFR and LIBOR correlation differences between tenor over the same date ranges[13]. At short tenors, the differences in correlation between observed and backfilled periods are most pronounced, peaking at 0.07 for 1-month SOFR and EFFR compared to 0.01 for EFFR and LIBOR at the same tenor. For tenors 6 months and beyond, the differences in SOFR and LIBOR correlations exceeds that between SOFR and EFFR. Backfilled SOFR at the 1-, 2-, and 3-month tenors exhibits less co-movement with proxy rates than actual SOFR does, indicating there is room for improvement. The

[12] See the Appendix for spreads.

[13] If a structural break did occur before 2018, it would show up in the EFFR and LIBOR correlation as well. There is no such divergence.

minimal difference in EFFR and LIBOR correlation between periods at each tenors shows the issue is the quality of backfilled SOFR, not a structural break affecting the co-movement between reference rates.

SOFR vs. Proxy Correlation Differences, Historical - Backfilled

Tenor	$\rho^{SOFR,EFFR}{}_{\tau,t\geq t0} - \rho^{SOFR,EFFR}{}_{\tau,t<t0}$	$\rho^{SOFR,LIBOR}{}_{\tau,t\geq t0} - \rho^{SOFR,LIBOR}{}_{\tau,t<t0}$	$\rho^{EFFR,LIBOR}{}_{\tau,t\geq t0} - \rho^{EFFR,LIBOR}{}_{\tau,t<t0}$
1M	0.069974	0.066537	0.011143
2M	0.048575	0.043026	0.005787
3M	0.034251	0.027013	0.003944
6M	0.007610	0.012943	0.011501
9M	0.002548	0.012818	0.012174
12M	0.002074	0.014091	0.011563
18M	0.001877	0.012485	0.009215
24M	0.003493	0.012298	0.007472
36M	0.002435	0.007744	0.004954
48M	0.003385	0.007180	0.003598
60M	0.003876	0.006565	0.002745
84M	0.002633	0.003897	0.001776
120M	0.001710	0.002200	0.001123
180M	0.000847	0.001899	0.000858
240M	0.001056	0.002880	0.000853
300M	0.001225	0.003651	0.000886
360M	0.001320	0.004149	0.000900
480M	0.001628	0.005044	0.001013
600M	0.001800	-	-

Correlation between SOFR and EFFR per tenor from 09/04/18 – 12/31/2020 minus correlation from 01/03/07 – 9/3/18. Correlation between SOFR and LIBOR per tenor from 09/04/18 – 12/31/2020 minus correlation from 01/03/07 – 9/3/18. Positive value indicates greater observed historical correlation than backfilled correlation. LIBOR data for unavailable for 600-month tenor. See the Appendix for cross-tenor differences.

The MDD's dynamic spread methodology identifies EFFR as a suitable proxy for SOFR[14]. As a preliminary test for the plausibility of EFFR as a proxy and to complement the MDD's correlation analysis, I perform time series regressions using the Sept. 4th, 2018, forward observed historical data. First, a regression in levels, per tenor (N=19), of SOFR on EFFR. Second, on first-differenced data, where I include a constant to test for a time-varying trend[15] in levels. If EFFR is a perfect proxy, the constant estimate should be 0 and the coefficient on EFFR should be exactly 1 for both the levels and first-differenced regression.

$$\mathrm{SOFR}_t^\tau = \beta_0 + \beta_1 \mathrm{EFFR}_t^\tau + \epsilon_t$$

$$\mathrm{SOFR}_t^\tau - \mathrm{SOFR}_{t-1}^\tau = \alpha_0 + \alpha_1(\mathrm{EFFR}_t^\tau - \mathrm{EFFR}_{t-1}^\tau) + u_t$$

For the level regressions, at each tenor the coefficient estimate for EFFR is roughly 1 and significant at a 1% level. R^2s are nearly one. Constant estimates are statistically indistinguishable from 0 for the 1-month to 3-month tenors, but non-zero and significant at a 1% level for 9 months and higher. They are largest in magnitude for the long tenors, specifically 120 months and above. Combined with a EFFR coefficient of 1, negative constant estimates for a tenor indicate that the SOFR – EFFR spread is negative

[14] The suitability of EFFR as a proxy for SOFR doesn't prove the non-existence of suitable alternative proxies. EFFR is readily available at all tenors of interest for the period in question and has a similar economic interpretation to SOFR as a short-term reference rate.

[15] First differencing a levels regression removes the constant. Adding a constant to the first differenced regression implies a time-trend in levels. Instead of imposing the restriction of no time trend by excluding a constant from the specification, I test the assumption of no time trend. A statistically significant non-zero constant is evidence of such a trend.

on average. Spreads are near-zero on average for 36-month tenors and below in the 9/4/18 – 12/31/20 data. The first-differenced regression results show that at short tenors, ranging from 1 month to 3 months, much of the day-to-day variation in SOFR isn't attributable to variation in EFFR. For example, R^2 for the 1-month tenor is just 24%, leaving 76% of the daily change in SOFR unaccounted for. For 36 months and above, R^2's are nearly one. The change in EFFR explains nearly all of the change in SOFR for these tenors. The coefficient estimates for the change in EFFR are nearly 1 and significant at a 1% level. Constant estimates are 0 and not significant. There is no evidence of a time-varying trend in levels.

Time Series Regression Results

Model	$SOFR_t^\tau = \beta_0 + \beta_1 EFFR_t^\tau + \varepsilon_t$			$\Delta SOFR_t^\tau = \alpha_0 + \alpha_1 \Delta EFFR_t^\tau + u_t$		
Tenor	$\hat{\beta}_0$	$\hat{\beta}_1$	$R^2_{Adj.}$	$\hat{\alpha}_0$	$\hat{\alpha}_1$	$R^2_{Adj.}$
1M	-0.003 (0.005)	1.020*** (0.003)	0.996	-0.000 (0.002)	0.954*** (0.071)	0.240
2M	-0.002 (0.004)	1.019*** (0.002)	0.997	-0.000 (0.001)	0.972*** (0.060)	0.315
3M	-0.000 (0.003)	1.018*** (0.002)	0.998	-0.000 (0.001)	0.961*** (0.050)	0.394
6M	0.004** (0.002)	1.017*** (0.001)	0.999	-0.000 (0.001)	0.954*** (0.027)	0.687
9M	0.006*** (0.002)	1.015*** (0.001)	0.999	-0.000 (0.000)	0.962*** (0.017)	0.845
12M	0.007*** (0.002)	1.013*** (0.001)	0.999	-0.000 (0.000)	0.968*** (0.012)	0.915
18M	0.007*** (0.001)	1.012*** (0.001)	1.000	-0.000 (0.000)	0.978*** (0.008)	0.964
24M	0.007*** (0.001)	1.010*** (0.001)	1.000	-0.000 (0.000)	0.984*** (0.006)	0.980
36M	0.002* (0.001)	1.010*** (0.001)	1.000	-0.000 (0.000)	0.990*** (0.004)	0.989
48M	-0.003** (0.001)	1.009*** (0.001)	1.000	-0.000 (0.000)	0.994*** (0.003)	0.993
60M	-0.007*** (0.001)	1.009*** (0.001)	1.000	-0.000 (0.000)	0.996*** (0.003)	0.995
84M	-0.016*** (0.001)	1.010*** (0.001)	1.000	-0.000 (0.000)	0.998*** (0.003)	0.996
120M	-0.027*** (0.002)	1.010*** (0.001)	0.999	-0.000 (0.000)	1.000*** (0.003)	0.996
180M	-0.041*** (0.002)	1.012*** (0.001)	0.999	-0.000 (0.000)	1.002*** (0.003)	0.995
240M	-0.051*** (0.002)	1.013*** (0.001)	0.999	-0.000 (0.000)	1.002*** (0.003)	0.993
300M	-0.058*** (0.003)	1.015*** (0.001)	0.999	-0.000 (0.000)	1.003*** (0.004)	0.993
360M	-0.064*** (0.003)	1.016*** (0.002)	0.999	-0.000 (0.000)	1.003*** (0.004)	0.992
480M	-0.070*** (0.003)	1.017*** (0.002)	0.998	-0.000 (0.000)	1.003*** (0.004)	0.991

Regression of SOFR on EFFR, 9/4/18+ daily data. T=577 for levels regression, T=576 for first-differenced regression. Standard errors in parenthesis. Newey-West standard errors with $T^{1/4}$ maximum lags. *** indicates significance at 1% level, ** at 5%, * at a 10% level.

The relationship between EFFR and SOFR persists after first-differencing for the 6 month tenor and higher, and is most convincing for the 36 month tenor and higher. At short tenors, EFFR is a serviceable proxy for SOFR. At long tenors, it is a strong proxy for SOFR[16].

Backfilled SOFR's time series properties largely match those of observed historical SOFR. Prior to 9/4/18, EFFR in levels at the 1-month to 36-month tenors is stationary and is non-stationary at the 48M and higher tenors. Backfilled SOFR in levels is stationary at the 1-month to 24-month tenors and is non-stationary at higher tenors. Both series are non-stationary at all tenors after 9/4/18. SOFR and EFFR exhibit serial correlation after 9/4/18 and backfilled SOFR and EFFR do as well at all tenors. First-differenced EFFR is stationary prior to and after 9/4/18, as is SOFR and backfilled SOFR[17]. However, first-differenced EFFR pre-9/4/18 shows serial correlation from the 1-month to 60-month tenors and no serial correlation at 84 months and higher. Backfilled SOFR shows serial correlation at all tenors. Post-9/4/18 first-differenced EFFR and SOFR show serial correlation from 1-month to 24-month tenors, and no serial correlation at 36 months and higher. First-differenced backfilled SOFR has serial correlation at higher tenors despite not having any at the same tenors for observed SOFR. Although not decisive, this is another reason to consider improvements to backfilled SOFR. The change in SOFR and EFFR for tenors over 84 months from 9/4/18 resembles white noise, whereas the change in backfilled SOFR does not. Time series properties alone do not render backfilled SOFR implausible, however the much higher volatility, reduced correlation with other reference rates, and visual evidence provide ample motivation to improve the quality of backfilled SOFR, especially at the shortest tenors.

[16] Regression and correlation results are so convincing that they may obviate a spread of any kind when backfilling. A dynamic spread model can certainly result in a backfilled SOFR series that is worse than just backfilling SOFR at longer tenors with historical EFFR, equivalent to a zero-spread model. There may be other reasons to use a spread, but EFFR is a viable first-order proxy for SOFR.

[17] SOFR in the MDD is EFFR plus a dynamic spread. If observed SOFR is non-stationary, it cannot be modelled as the sum of a stationary proxy and stationary spread. The combination of stationary and non-stationary or non-stationary and non-stationary variables can be stationary, a phenomenon called cointegration. As a trivial example, $X(t) \sim N(\mu(t), \sigma^2(t))$ and $Y(t) \sim N(\mu(t), \sigma^2(t))$ are non-stationary, but $Z(t) = X(t) - Y(t)$ is weakly stationary with mean 0 and constant variance for $Cov(X(t),Y(t)) = \sigma^2(t)$. However, the sum of two processes with constant moments cannot produce a process with time-varying moments. EFFR is non-stationary at all tenors after 9/4/18, as is SOFR. The SOFR – EFFR spread empirically is stationary for short and long tenors in the 9/4/18 – 12/30/21 data and non-stationary for middle tenors, demonstrating the above point. Prior to 9/4/18, EFFR is stationary until the 36-month tenor and non-stationary for higher tenors. Under the ABM process, the spread has time-varying variance and so it should not be stationary at any tenor. Indeed, ADF tests of the pre-9/4/18 spread computed using backfilled SOFR confirm this is the case for all tenors except for 9M and 12M, which do have stationary spreads. Summing a non-stationary spread with EFFR that is stationary or non-stationary depending on the tenor pre-9/4/18 to backfill SOFR can plausibly result in stationary or non-stationary SOFR. ADF tests of the stationary of the backfilled SOFR series alone do not rule out the ABM methodology. However, if the spread is modelled as multivariate normal white noise, then it is stationary at every tenor, and tenors for which EFFR is also stationary should have stationary backfilled SOFR because backfilled SOFR at that tenor will be the sum of two stationary series. "Consistent" stationarity is thus a necessary but insufficient test to identify a plausible backfilling methodology.

Revised Testing Framework

To validate the methodology for backfilling SOFR, the MDD compares actual observed SOFR from Sept. 2018 onward to SOFR predicted over the same period by using the backfilling methodology to forward fill. Comparisons involve summary statistics or properties and functions of the series like VaR. There are several problems. First, comparisons of correlation across tenors, summary statistics, or volatility measures aren't specific enough to assess *relative* performance. The results of such comparisons all beg the question: compared to what? EFFR with no spread may perform just as well, if not better, than fitted SOFR in such comparisons. A forward filled SOFR series that has a completely different level can have the same correlation, and so forth. Second, it constitutes in sample testing. When fitted SOFR is compared to actual observed SOFR over the same period in which those models are calibrated, it is easy to overstate performance. A proper evaluation framework must feature a separate testing sample, which contains observations the model has not been exposed to during the training or validation process. Third, the test for the plausibility of backfilled SOFR cannot exclusively be SOFR predicted from 2018 onward compared to actual SOFR from 2018 onward. SOFR is a market-based reference rate and it would strain credulity if it were to diverge wildly from EFFR or LIBOR in the backfilled period. In a subsequent section, I address the issue of the lack of SOFR observations pre-2018. Finally, recomputing VaR or SVaR on the forward filled series and comparing it to VaR or SvaR computed on the historical series isn't meaningful. It is unclear whether either metric is sensitive enough to the particular time series to show a non-random difference. Recomputing VaR on the same time series, for example, can produce a different figure due to the nature of random draws. Furthermore, this test cannot differentiate backfilling methodologies that have the same magnitude of VaR change but different signs. I outline a new process to test the performance of backfilling methodologies. Per the MDD, the period of observed historical SOFR is Sept. 4th, 2018 – Dec. 31st, 2020. I temporally partition this period into training, validation, and test sets, comprised of 70%, 10%, and 20% of the data, respectively. The temporal split preserves the time series structure of the data[18]. Because the goal is to predict backwards in time, the test period precedes the validation period, which in turn precedes the training period.

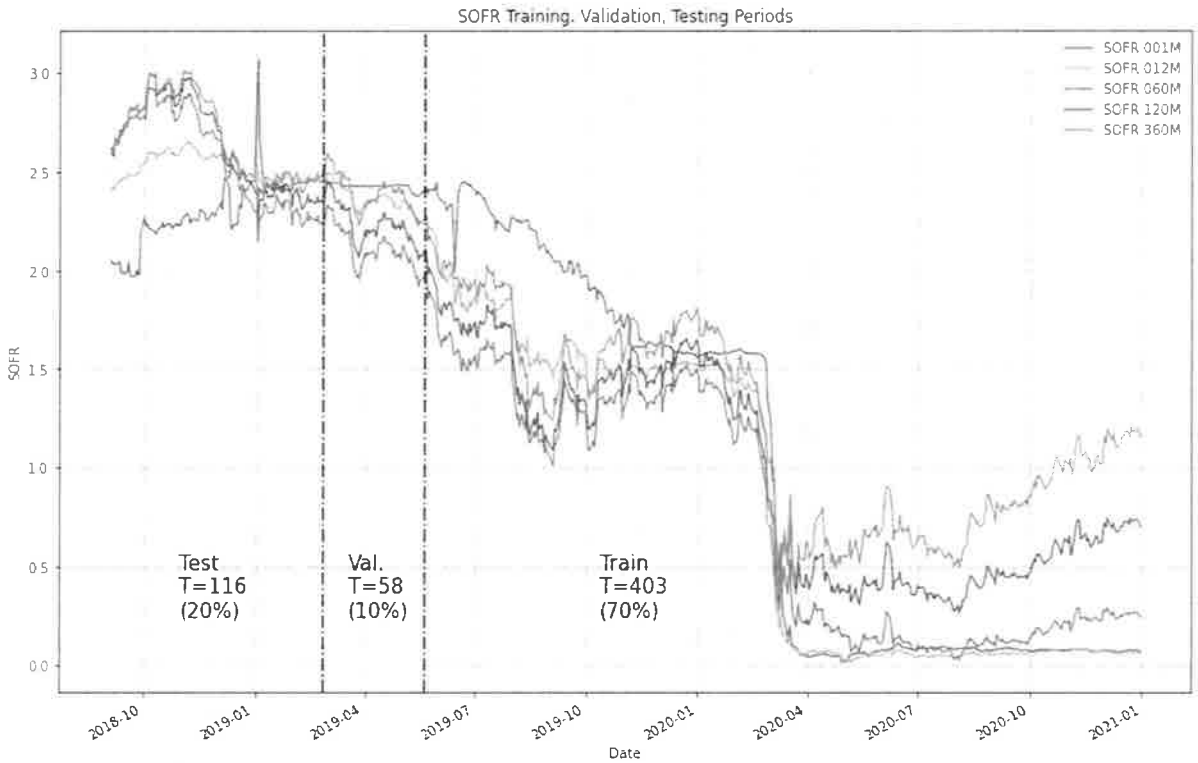
Retesting Period Setup

Period	Range	T	% of Sample
Train	2019-05-22 – 2020-12-31	403	69.84%
Validation	2019-02-26 – 2019-05-21	58	10.05%
Test	2018-09-04 – 2019-02-25	116	20.10%

Date ranges and sample percentages for temporal training, validation, and testing periods. Test precedes validation which precedes training.

Principal components regression requires a validation set to select the optimal number of principal components. Other backfilling methodologies don't require hyperparameter tuning, and for those I combine the training and validation sets to train the models on a full 80% of the sample. Samples are disjoint by construction. A key concern for time series data is the comparability of the training, validation, and testing periods. In a time series, the levels, trend, and volatility can easily vary across samples. Specifically, the concern is that the relationship between time series predictors and response breaks across samples because the time series properties of the response variable differ. This concern is not present in cross-sectional data but the cost of a cross-sectional approach is the destruction of all time series properties of the sample, which is worse. I evaluate the comparability of the sample periods.

[18] A random split of the data not only destroys all time series properties but can greatly overstate model performance. A model that observes training data at t_1 and t_3 can simply interpolate to predict t_2. In effect, the model just fills in the training period's trend on the test set.



Time series of 1-, 12-, 60-, 120-, and 360-month SOFR to illustrate test (2018-09-04 – 2019-02-25, T= 116, 20.10%), validation (2019-02-26 – 2019-05-21, T=58, 10.05%), and train (2019-05-22 – 2020-12-31, T=403, 69.84%) periods.

SOFR trends downward in the observation period for all tenors, as evident in the graph above. Summary statistics (see the Appendix) confirm that the average and median SOFR across the tenors are most alike in the validation and test samples, with shorter tenors tending to have more similar sample moments, although the volatility for the testing period's shortest tenors is considerably higher than the validation period. The discrepancy between testing and training average and median SOFR is also off by a factor of roughly 10 for short tenors, and roughly 2 for longer tenors. Based upon the summary statistics, the temporally delineated samples are in fact quite different. Despite this issue, the temporal training and testing approach is still superior to the alternative. First, the covariates for each model are EFFR at various tenors, which show similar time series trends and summary statistics as SOFR in each period. The sample moments of the dependent variable differ across sample splits, but the sample moments of the independent variables differ in the exact same regard. Second, the alternative of a randomized split is worse. Such a training and testing configuration is likely to overstate performance, whereas the temporal configuration, if biases test and validation error at all, is likely to understate performance. A different temporal split, such as 45/10/45, leaves insufficient training data. Third, the MDD's approach of comparing forward filled SOFR's summary statistics and functions thereof to observed SOFR's summary statistics and functions thereof does not include a single explicit test error measure, which is untenable. Separating the observed period into test, validation, and train is superior.

Model Estimation and Performance

For each model outlined earlier, I fit the model on the 5/22/19 – 12/31/2020 training set, fine tune hyperparameters on the 2/26/19 – 5/21/19 validation set, predict over the 9/4/18 – 2/25/19 test set, and compute regression error measures by comparing predicted SOFR values to actual SOFR values per tenor. For models that do not have hyperparameters, the training set is 2/26/19 – 12/31/2020. Spread models predict the SOFR-EFFR spread per tenor, so I add the spread to EFFR to get predicted SOFR in the validation and test periods. I outline estimation steps and performance below. I test five models. Two, the arithmetic Brownian motion process and multivariate normal white noise process, predict a dynamic SOFR – EFFR spread per tenor. Another two, an average and zero-spread model, predict a constant SOFR – EFFR spread per tenor. Principal components regression predicts SOFR itself per tenor.

Arithmetic Brownian Motion (MDD)

First, I test the arithmetic Brownian motion process to backfill spreads. Although the use of multivariate normal draws to generate cross-tenor correlation in backfilled spreads as prescribed in the MDD is imprecise, I replicate the methodology exactly as it is outlined in the MDD. First, I calibrate the covariance matrix for the multivariate normal distribution of the change in tenor spreads by taking the sample covariance of the daily change in spreads during the training period. The vector of mean changes is zero by assumption. Second, I draw from the multivariate normal distribution for each tenor and date in the training and testing periods to get simulated changes in spreads. The first temporal observation in the training set is 12/30/21, for which there is no subsequent spread to use as the initial spread, so I replace the drawn changes in spread for that day with 0s for all tenors. Third, I multiply each time series of draws (one vector per tenor) by the corresponding fat-tail scaled volatility, which I computed earlier on the observed spread training data. Fourth, I take the cumulative sum of scaled spreads per tenor in reverse temporal order to obtain summed changes in spreads at each day per tenor. Fifth, I subtract these cumulative sums of changes from the initial observed spread, which for the training set is 12/30/21 and for the test set is 5/22/2019. The result is a series of backfilled SOFR – EFFR spreads per tenor for the training period of 2/26/2019 – 12/31/2020 and testing period of 9/4/2018 – 2/25/2019, which I compare to observed historical spreads over the same period to generate spread error measures. Sixth, I add EFFR to the corresponding backfilled spread per tenor to generate backfilled SOFR for the training and testing periods. I compare these backfilled SOFR series to the observed historical SOFR to generate SOFR error measures. Seventh, I vary the random seed used to generate draws from the multivariate normal distribution and repeat the aforementioned steps, collecting the test R^2 and RMSE values for the spreads and SOFR for each seed to create a distribution of error measures.

This process has two advantages. It tests the quality of the spreads, thus isolating one source of error. Testing backfilled SOFR alone is insufficient because the correlation and similarity in levels of EFFR and SOFR masks the distortion from spreads that are inaccurate but small in magnitude. In addition, backfilled spreads will vary considerably with the random draw because of the sequential summing of drawn spreads, so error metrics based on a single draw are woefully insufficient[19]. I report train and test spread error measures per tenor for a fixed random seed, and then summary statistics for test errors. Backfilled spreads show large negative training R^2s, with mean absolute percentage errors well over 100%. Training spreads are worse than testing period spreads despite fat-tail scaled volatility and cross-tenor correlations calibrated to the training period because the spread variance increases considerably over the much longer training period. Test R^2 is positive at 0.006 only for the 1-month spread. The

[19] For intuition, consider estimating a population mean with a random sample of size 1. The sample mean with n=1 would have the same variance as the entire population. Repeating the testing process many times and varying the random seed used to generate random draws each time is an approach to generate non-parametric standard errors for an estimator, in this case the sample mean of a test error measure. Each random seed r = 1…1,000 has an associated test error measure, which I can average and compute a standard error. See the Appendix.

lowest training MAPE is 83.5% for the 600-month spread. 1-month training MAPE is an astonishing 12,600%. The 1-month SOFR-EFFR spread is on average 2.4% from 9/4/18 – 12/30/21, but predicted spreads deviate by 5.6 percentage points on average, a result of sample end outlier predicted spreads.

ABM Backfilled Spread Error Measures

Spread	Train				Test			
Tenor	R^2	MAPE	MSE	RMSE	R^2	MAPE	MSE	RMSE
001M	-0.241	126.169	0.003	0.056	0.006	6.611	0.012	0.110
002M	-0.317	6.138	0.002	0.048	-0.016	2.017	0.008	0.092
003M	-0.496	5.379	0.002	0.048	-0.103	1.254	0.005	0.072
006M	-0.650	4.871	0.002	0.044	-0.609	2.073	0.002	0.044
009M	-1.473	37.190	0.003	0.050	-1.078	2.842	0.001	0.038
012M	-1.905	6.237	0.002	0.049	-1.380	4.892	0.001	0.035
018M	-2.036	4.178	0.002	0.044	-0.863	3.543	0.001	0.025
024M	-2.105	13.985	0.002	0.040	-0.292	2.295	0.000	0.019
036M	-2.140	9.998	0.001	0.035	-0.115	2.701	0.000	0.017
048M	-2.316	8.596	0.001	0.032	-0.658	1.297	0.001	0.023
060M	-2.450	9.299	0.001	0.031	-1.139	1.386	0.001	0.028
084M	-2.502	4.852	0.001	0.030	-1.388	3.764	0.001	0.036
120M	-2.798	3.523	0.001	0.032	-1.027	1.467	0.002	0.045
180M	-3.174	8.187	0.001	0.033	-0.614	3.859	0.003	0.054
240M	-3.225	1.710	0.001	0.033	-0.486	1.797	0.003	0.059
300M	-3.183	1.169	0.001	0.033	-0.421	1.788	0.004	0.062
360M	-3.004	0.991	0.001	0.034	-0.383	1.811	0.004	0.065
480M	-2.762	0.875	0.001	0.035	-0.338	1.852	0.005	0.068
600M	-2.558	0.835	0.001	0.036	-0.315	1.879	0.005	0.070

Training and test period R^2, root mean-squared error, mean squared error, and mean absolute percentage error for backfilled SOFR-EFFR spreads against observed SOFR-EFFR spreads per tenor. Training period covers 2/26/2019 – 12/31/2020 (T=461) and testing period of 9/4/2018 – 2/25/2019 (T=116). Spreads follow an arithmetic Brownian motion process constructed per the MDD. Measures reported for seed 42.

Negative R^2s indicates that just using the in sample spread average outperforms backfilled spreads using the ABM process. Overall, the ABM's dynamic spreads at each tenor have minimal relation to observed spreads per tenor. This is not unexpected, as Brownian motion stochastic processes are not used to generate accurate predictions. As stated earlier, the ABM generates dynamic spreads for the sake of having time-varying spreads, but otherwise serves little purpose. Despite poor spreads, backfilled SOFR performs well, especially at higher tenors.

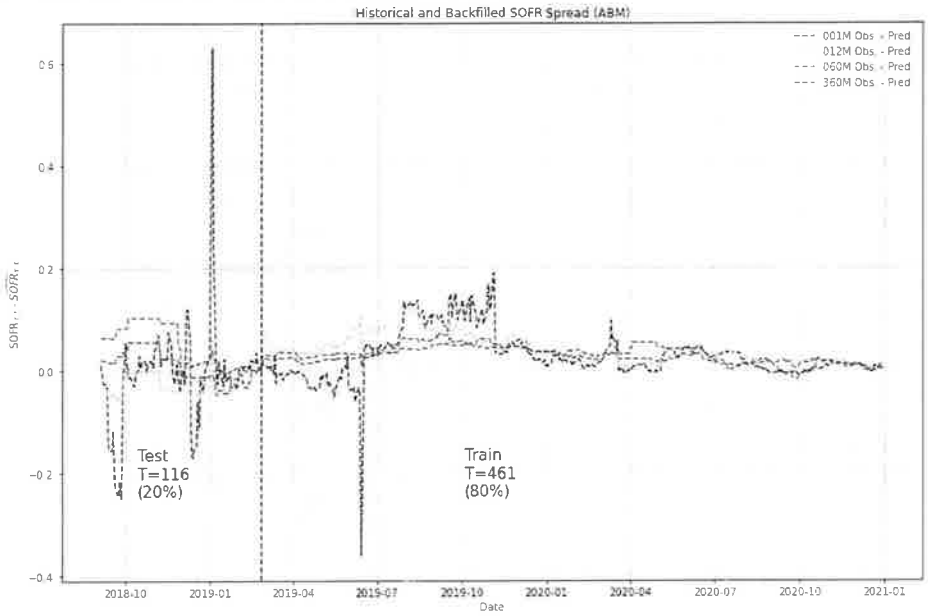
ABM Backfilled SOFR Error Measures

Spread	Train				Test			
Tenor	R^2	MAPE	MSE	RMSE	R^2	MAPE	MSE	RMSE
001M	0.997	0.103	0.003	0.056	0.623	0.026	0.012	0.110
002M	0.998	0.103	0.002	0.048	0.643	0.022	0.008	0.092
003M	0.998	0.113	0.002	0.048	0.691	0.017	0.005	0.072
006M	0.998	0.114	0.002	0.044	0.655	0.014	0.002	0.044
009M	0.997	0.157	0.003	0.050	0.498	0.012	0.001	0.038
012M	0.997	0.179	0.002	0.049	0.713	0.011	0.001	0.035
018M	0.997	0.218	0.002	0.044	0.954	0.008	0.001	0.025
024M	0.998	0.542	0.002	0.040	0.987	0.006	0.000	0.019

036M	0.998	0.917	0.001	0.035	0.994	0.004	0.000	0.017
048M	0.998	0.133	0.001	0.032	0.990	0.006	0.001	0.023
060M	0.998	0.061	0.001	0.031	0.985	0.008	0.001	0.028
084M	0.998	0.038	0.001	0.030	0.976	0.010	0.001	0.036
120M	0.998	0.031	0.001	0.032	0.961	0.012	0.002	0.045
180M	0.997	0.028	0.001	0.033	0.939	0.015	0.003	0.054
240M	0.997	0.026	0.001	0.033	0.924	0.017	0.003	0.059
300M	0.997	0.026	0.001	0.033	0.913	0.018	0.004	0.062
360M	0.997	0.027	0.001	0.034	0.903	0.019	0.004	0.065
480M	0.997	0.030	0.001	0.035	0.892	0.021	0.005	0.068
600M	0.997	0.034	0.001	0.036	0.884	0.022	0.005	0.070

Training and test period R^2, root mean-squared error, mean squared error, and mean absolute percentage error for backfilled SOFR against observed SOFR per tenor. Backfilled SOFR equals EFFR plus backfilled spread per tenor. Training period covers 2/26/2019 – 12/31/2020 (T=461) and testing period of 9/4/2018 – 2/25/2019 (T=116). Spreads follow an arithmetic Brownian motion process constructed per the MDD. Measures reported for spread generated with random seed 42.

Training R^2s are nearly 1 for every tenor, reflecting the strong co-movement between EFFR and SOFR in the training period. RMSEs are nearly identical to those of the backfilled spreads, confirming that the spreads are the primary driver of the SOFR error. Test R^2s are lower than training for every tenor. RMSE is lower as well for mid tenors, such as the 9-, 12-, 18-, 24-, 36-, 48-, and 60-month tenor. 9-month backfilled SOFR has the lowest test R^2 at 0.498, and test R^2 is mostly increasing in tenor, peaking at 0.994 for 36 months. Spreads do not exhibit a similar pattern, suggesting that improved test R^2 for SOFR at higher tenors is due to more co-movement between SOFR and EFFR instead of more accurate spreads.



$SOFR_t^\tau$ - $\widehat{SOFR}_t^\tau$ for 1-, 12-, 60-, and 360-month tenors using arithmetic Brownian motion process with random seed 42. Vertical line denotes split between training period (2/26/2019 – 12/31/2020, T=461) and testing period of (9/4/2018 – 2/25/2019, T=116).

I plot the spread between observed SOFR and backfilled SOFR produced using the ABM process for four tenors. The 1-month backfilled SOFR series shows the most deviation from observed SOFR. The arithmetic Brownian motion approach produces plausible backfilled SOFR because the proxy rate, EFFR, co-moves with SOFR and has a similar enough level that even inaccurate spreads do not sufficiently distort the imputed values. Average spread test R^2s from varying the random seed are higher than those

reported above, but are all still negative. For short tenors – up to 18 months - average SOFR test R^2s are lower than those reported above, but for 18 months and higher the average SOFR test R^2 and RMSEs are a match. Short tenor spreads, and thus backfilled SOFR, are highly susceptible to variation in random draws from the multivariate normal distribution, but mid and long tenors are not. Backfilled SOFR for 1-month to 12-months can be widely different over repeated runs.

White Noise Model

A white noise model produces dynamic spreads by drawing from a multivariate normal distribution that exhibits no serial correlation across draws but shows cross-tenor correlation within a vector of draws. The moments of the white noise's multivariate normal distribution are calibrated using historical spreads, not the historical change in daily spreads. Further, I don't impose the restriction of a 0 mean vector as done in the MDD's arithmetic Brownian motion process. I fit the white noise model as follows. First, I compute the sample mean and covariance matrices using the training period's SOFR – EFFR spreads. The covariance matrix captures cross-tenor correlation. Second, I take independent draws of spread vectors from the calibrated distribution. These are the predicted spreads in the test period. Third, I add the predicted spreads to observed EFFR in the test period at the corresponding tenors, resulting in backfilled SOFR for the test period. Fourth, I repeat the above process, but vary the random seed, collecting spread and SOFR test error measures at each tenor for each random seed, and examine the empirical distributions of error measures. Step four shows the sensitivity of test errors to random draws. For each tenor, there is no imposed correlation structure on the time series of backfilled spreads, which in theory avoids the issue of a spread's variance increasing mechanically as a function of time.

White Noise Test Error

Tenor	Backfilled SOFR – EFFR Spread				Backfilled SOFR			
	R^2	MAPE	MSE	RMSE	R^2	MAPE	MSE	RMSE
001M	-0.245	7.980	0.015	0.123	0.528	0.036	0.015	0.123
002M	-0.233	3.197	0.010	0.102	0.567	0.029	0.010	0.102
003M	-0.353	1.633	0.006	0.080	0.620	0.023	0.006	0.080
006M	-1.128	1.874	0.003	0.051	0.544	0.016	0.003	0.051
009M	-1.709	2.252	0.002	0.043	0.346	0.014	0.002	0.043
012M	-1.980	3.839	0.002	0.039	0.641	0.012	0.002	0.039
018M	-2.242	3.310	0.001	0.033	0.920	0.011	0.001	0.033
024M	-2.254	2.473	0.001	0.030	0.968	0.009	0.001	0.030
036M	-1.735	4.570	0.001	0.027	0.984	0.008	0.001	0.027
048M	-1.368	4.650	0.001	0.027	0.986	0.008	0.001	0.027
060M	-1.201	2.313	0.001	0.029	0.985	0.008	0.001	0.029
084M	-0.956	3.683	0.001	0.033	0.980	0.010	0.001	0.033
120M	-0.553	1.295	0.002	0.039	0.970	0.011	0.002	0.039
180M	-0.300	4.043	0.002	0.048	0.951	0.014	0.002	0.048
240M	-0.255	1.628	0.003	0.054	0.936	0.016	0.003	0.054
300M	-0.249	1.626	0.003	0.059	0.923	0.018	0.003	0.059
360M	-0.252	1.667	0.004	0.062	0.913	0.019	0.004	0.062
480M	-0.259	1.731	0.004	0.066	0.898	0.021	0.004	0.066
600M	-0.267	1.779	0.005	0.068	0.888	0.022	0.005	0.068

Training and test R^2, root mean-squared, mean squared, and mean absolute percentage error for backfilled SOFR against observed SOFR. Backfilled SOFR equals EFFR plus backfilled spread per tenor. Training period covers 2/26/2019 – 12/31/2020 (T=461) and testing period of 9/4/2018 – 2/25/2019 (T=116). Spreads follows a multivariate normal distributed white noise process. Measures reported for random seed 42.

The predicted spreads of the white noise model are less accurate in the test period than those of the ABM process for the 1-month to 60-month tenors, but more accurate for the 84-month to 600-month tenors, as evidenced by higher (but still negative) R^2s and lower RMSEs for the longer tenors. Backfilled SOFR from the white noise model exhibits the same pattern. Relative to the white noise process, the ABM process has better test performance for short tenors but worse test performance for long tenors. The above table shows error measures for the same random seed used to generate draws and thus compute error measures for the ABM process. The same pattern manifests when comparing averages of white noise test error measures for the backfilled spread and SOFR to ABM averages; the ABM process shows superior performance for shorter tenors, and the white noise superior for longer tenors.

Average Spread Model

A constant spread model sets backfilled SOFR equal to EFFR plus a fixed spread per tenor, eliminating model dynamics as a source of variation in the spread and the backfilled SOFR. I compute the average spread per tenor during the training period and then add it to the corresponding EFFR series during the test period to get backfilled SOFR, using the same training and testing periods of 2/26/2019 – 12/31/2020 and 9/4/2018 – 2/25/2019, respectively[20]. I present test error measures for the backfilled spreads and SOFR below. Unsurprisingly, spread test R^2s for short tenors are nearly 0 and range from -0.130 to -0.516 for longer tenors. No spread test R^2s are positive, but at all tenors they are higher than the ABM's except for the 1-month tenor (-0.005 for the average model versus 0.006 for the ABM process). More importantly, the average model's test R^2s are higher than the average of the ABM test R^2s computed from varying the random seed at all tenors except 36 months. The average model's test RMSEs are lower or similar to the average of ABM test RMSEs computed from varying the random seed, with comparable figures for the 120-month tenors and higher. Test SOFR R^2 and RMSE for the average spread model shows a similar pattern, outperforming the ABM process for all tenors. The average spread model outperforms the ABM spread model at minimal computational cost.

Average Spread Model Test Error

Tenor	Backfilled SOFR – EFFR Spread				Backfilled SOFR			
	R^2	MAPE	MSE	RMSE	R^2	MAPE	MSE	RMSE
001M	-0.005	5.852	0.012	0.110	0.619	0.026	0.012	0.110
002M	-0.001	1.594	0.008	0.092	0.648	0.022	0.008	0.092
003M	-0.001	0.777	0.005	0.069	0.719	0.015	0.005	0.069
006M	-0.006	1.084	0.001	0.035	0.784	0.010	0.001	0.035
009M	-0.011	1.469	0.001	0.027	0.756	0.009	0.001	0.027
012M	-0.010	2.452	0.001	0.023	0.878	0.008	0.001	0.023
018M	-0.000	1.960	0.000	0.019	0.975	0.007	0.000	0.019
024M	-0.009	1.429	0.000	0.016	0.990	0.005	0.000	0.016
036M	-0.130	2.667	0.000	0.017	0.994	0.004	0.000	0.017
048M	-0.317	2.372	0.000	0.020	0.992	0.005	0.000	0.020
060M	-0.454	0.974	0.001	0.023	0.990	0.006	0.001	0.023
084M	-0.516	1.833	0.001	0.029	0.985	0.008	0.001	0.029
120M	-0.354	1.126	0.001	0.037	0.974	0.011	0.001	0.037
180M	-0.231	2.914	0.002	0.047	0.953	0.015	0.002	0.047
240M	-0.209	1.582	0.003	0.053	0.938	0.017	0.003	0.053
300M	-0.210	1.618	0.003	0.058	0.926	0.018	0.003	0.058

[20] A 50/50 training/test split is feasible, but I reuse the existing setup to ensure test error measures are comparable across methodologies.

360M	-0.215	1.669	0.004	0.061	0.915	0.019	0.004	0.061
480M	-0.218	1.741	0.004	0.065	0.901	0.021	0.004	0.065
600M	-0.221	1.789	0.004	0.067	0.892	0.022	0.004	0.067

Test period R^2, root mean-squared error, mean squared error, and mean absolute percentage error for backfilled SOFR against observed SOFR per tenor. Backfilled SOFR equals EFFR plus backfilled spread per tenor. Training period covers 2/26/2019 – 12/31/2020 (T=461) and testing period of 9/4/2018 – 2/25/2019 (T=116). Spreads equal training period average.

The average spread model similarly outperforms the white noise process model, with higher test R^2 for spreads and SOFR and lower RMSE at all tenors. Superior performance and minimal model risk and computational cost are arguments in favor of the average spread.

Zero Spread Model

A logical extension of the constant spread model is a spread of 0, meaning backfilled SOFR at each tenor just equals EFFR at the corresponding tenor. This model is the baseline for performance and has no computational cost[21]. I present error measures for SOFR using the same training and testing periods.

0 Spread Model Test Error (Backfilled SOFR vs. Observed SOFR)

Tenor	Training				Testing			
	R^2	MAPE	MSE	RMSE	R^2	MAPE	MSE	RMSE
001M	0.997	0.102	0.003	0.056	0.610	0.029	0.012	0.112
002M	0.998	0.086	0.002	0.048	0.631	0.025	0.009	0.094
003M	0.998	0.077	0.002	0.046	0.689	0.019	0.005	0.073
006M	0.998	0.079	0.002	0.044	0.685	0.013	0.002	0.042
009M	0.998	0.083	0.002	0.041	0.569	0.011	0.001	0.035
012M	0.998	0.082	0.001	0.038	0.761	0.010	0.001	0.032
018M	0.998	0.087	0.001	0.033	0.942	0.009	0.001	0.028
024M	0.999	0.135	0.001	0.030	0.975	0.008	0.001	0.026
036M	0.999	0.322	0.001	0.023	0.987	0.007	0.001	0.025
048M	0.999	0.099	0.000	0.019	0.989	0.007	0.001	0.024
060M	0.999	0.045	0.000	0.017	0.989	0.007	0.001	0.025
084M	0.999	0.031	0.000	0.017	0.988	0.008	0.001	0.026
120M	0.999	0.029	0.000	0.022	0.980	0.011	0.001	0.032
180M	0.998	0.031	0.001	0.031	0.961	0.015	0.002	0.043
240M	0.996	0.035	0.001	0.038	0.946	0.017	0.002	0.050
300M	0.995	0.039	0.002	0.043	0.933	0.019	0.003	0.055
360M	0.994	0.042	0.002	0.046	0.923	0.020	0.003	0.058
480M	0.993	0.049	0.003	0.051	0.909	0.022	0.004	0.062
600M	0.992	0.057	0.003	0.054	0.899	0.023	0.004	0.065

Test period R^2, root mean-squared error, mean squared error, and mean absolute percentage error for backfilled SOFR against observed SOFR per tenor. Backfilled SOFR equals EFFR. Training period: 2/26/2019 – 12/31/2020 (T=461), and testing period: 9/4/2018 – 2/25/2019 (T=116).

Spread models – dynamic or constant – obfuscate the source of the discrepancy between backfilled SOFR and observed SOFR, as the imputed spread and the variability of the proxy rate both contribute. A zero spread model identifies error attributable solely to the proxy rate. Training R^2s are nearly 1, indicating EFFR at all tenors exhibits near perfect co-movement with SOFR. Test R^2s are modest for short tenors, with a low of 0.569 at 9 months, but higher for 18-month and above tenors. SOFR exhibited

[21] Backfilled SOFR equals EFFR, so the SOFR-EFFR spread is always 0. This may pose challenges for instruments linked to the spread between the two rates or elements of the VaR engine that assume a non-zero spread.

more volatility at the onset of its usage, which coincides with the range of the test sample, hence there is lower co-movement for shorter tenors in the testing period. The average test R^2s for backfilled SOFR from the ABM process are lower than those of the 0 spread model for all tenors except the 18-month (0.013) , 24-month (0.012), 36-month (0.007), and 48-month (0.001) tenors. Standard errors for the average test R^2s are near zero for those tenors. The differences indicate that the ABM model either worsens backfilled SOFR by distorting the SOFR – EFFR spread or has a negligible but still observable negative effect. RMSEs show the same pattern. The zero spread model's test R^2 exceeds the average test R^2 for the white noise model at all tenors, although for some tenors the differences are marginal. Compared to the constant average spread model, the 0 spread model performs worse at the 1-month to 84-month tenors, and better at the 120-month to 600-month tenors. This also likely reflects the volatility of early SOFR, which is more pronounced at the shorter tenors.

PCA Model

The principal components regression model backfills SOFR in levels, as opposed to backfilling a spread added to observed EFFR. PCA produces a low-dimensional cross-tenor factor structure from observed effective federal funds rates, and principal components regression models observed SOFR as a linear combination of those factors by regressing SOFR on the principal component scores. I estimate the model as follows[22]. First, perform PCA on the 2019-05-22 – 2020-12-31 training set of all EFFR tenors. PCA is unsupervised, so there are no inputs to the model except for the panel of covariates. Second, compute the principal component scores for all principal components. For K covariates, PCA produces K principal components, but the goal is to select M principal components, M << K, that explain most of the variance in the original K covariates. Third, for each tenor, regress training period SOFR on M principal component scores and store the estimated coefficients. Each tenor has its own regression and estimated coefficients. Fourth, apply the loadings used to compute the principal component scores in the training period to the set of EFFR tenors in the validation period, 2019-02-26 – 2019-05-21, and obtain validation period principal component scores. Fifth, take a linear combination of validation period scores and coefficients estimated in the training period to obtain validation period SOFR per tenor. Sixth, compute error metrics by comparing validation period backfilled SOFR to validation period observed SOFR. Seventh, repeat steps 1 – 6, varying the number of principal components, M, used in the regression and selecting the M* that maximizes validation period[23] adjusted R^2. I vary M over the narrow range of 1 to 5, inclusive, consistent with the goal of obtaining a lower-dimensional representation, and use R^2_{Adj} to penalize additional covariates. Each tenor has its own M*. Finally, for each tenor, using the optimal M* principal component model as determined by the validation set, compute the testing period, 2018-09-04 – 2019-02-25, principal component scores using the training period loadings and observed EFFR and backfilled SOFR using the training period estimated coefficients. I report error measures from all periods – training, validation, test – for each tenor's optimal PCR model[24].

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[22] PCR in Sci-Kit Learn is fit like any other regression model, obviating separate estimation of principal component loadings and computation of principal component scores. In one step, I perform PCA and regress a dependent variable on a select number of principal component scores.

[23] The presence of a hyperparameter necessitates a validation set of some form. Cross-validation using randomized subsets of the training data, such as K-fold or LOOCV, has the same issues as a randomized train-test split. It is likely to overstate performance by picking up on trends. For time series data, a temporally separate validation period is critical to fine tune any hyperparameters.

[24] Selecting M* based upon test error is akin to manually selecting covariates in a regression based upon test error. It negates the purpose of a separate, completely unseen test set to assess model performance. The validation set and test set have no overlap precisely so that I can compare test error performance across measures. A reasonable concern is that the validation set, with 58 observations (10%), is small enough to produce noisy validation error measures, meaning M* may actually be suboptimal. However, to ensure comparability of test error performance across models, I cannot move more data from the test period into the validation period. To ensure stable coefficients estimated using PCR, moving training period data into the validation period is also problematic. A 50/50 split of the training period into training and validation, for example, exposes the PCR model to considerably less training data than the other models. As a sensitivity check, I examined the PCR validation period errors per SOFR tenor and I am satisfied with the margin between best and second best performing model.

Below are the optimal number of principal components per tenor and error measures for each period. PCR performs poorly in the test set at short tenors, with near-zero and negative test R^2 for the 6-month and 9-month tenors, respectively. Poor performance below 12 months may be a consequence of short-tenor volatility and the resulting breakdown in co-movement with the longer EFFR tenors. By 36 months, test R^2 exceeds 0.95 but declines to 0.926 after 300 months. Errors are generally increasing from training to validation to test periods. The modest test MAPE, of at most 4.2% for the 1-month, indicates the levels of predicted SOFR are correct. PCR performs best for the mid-tenors.

PCR Results

Tenor	M*	Train				Validation				Test			
		R^2	MAPE	MSE	RMSE	R^2	MAPE	MSE	RMSE	R^2	MAPE	MSE	RMSE
1M	5	0.999	0.138	0.001	0.034	0.371	0.005	0.000	0.014	0.356	0.042	0.021	0.144
2M	5	0.999	0.101	0.001	0.028	-0.91	0.010	0.001	0.026	0.389	0.037	0.015	0.121
3M	4	0.999	0.068	0.001	0.026	-2.84	0.014	0.001	0.036	0.642	0.021	0.006	0.078
6M	3	0.999	0.116	0.001	0.022	0.566	0.007	0.000	0.020	0.056	0.027	0.005	0.073
9M	2	0.999	0.162	0.001	0.024	0.969	0.003	0.000	0.009	-0.17	0.020	0.003	0.058
12M	5	1.000	0.089	0.000	0.016	0.965	0.004	0.000	0.013	0.869	0.007	0.001	0.023
18M	5	1.000	0.084	0.000	0.012	0.920	0.010	0.001	0.025	0.847	0.016	0.002	0.046
24M	5	1.000	0.127	0.000	0.011	0.983	0.005	0.000	0.014	0.949	0.011	0.001	0.037
36M	5	1.000	0.477	0.000	0.009	0.966	0.009	0.000	0.020	0.988	0.009	0.001	0.024
48M	4	0.999	0.131	0.000	0.014	0.908	0.015	0.001	0.032	0.982	0.011	0.001	0.031
60M	4	1.000	0.039	0.000	0.012	0.891	0.016	0.001	0.034	0.984	0.011	0.001	0.030
84M	4	1.000	0.018	0.000	0.011	0.947	0.010	0.001	0.023	0.980	0.011	0.001	0.033
120M	5	1.000	0.012	0.000	0.011	0.987	0.004	0.000	0.012	0.992	0.007	0.000	0.021
180M	5	1.000	0.009	0.000	0.009	0.992	0.003	0.000	0.009	0.977	0.011	0.001	0.033
240M	4	1.000	0.008	0.000	0.009	0.973	0.006	0.000	0.016	0.959	0.015	0.002	0.043
300M	3	1.000	0.010	0.000	0.010	0.961	0.008	0.000	0.019	0.949	0.017	0.002	0.048
360M	3	0.999	0.011	0.000	0.012	0.972	0.006	0.000	0.016	0.944	0.017	0.002	0.049
480M	3	0.999	0.015	0.000	0.015	0.987	0.004	0.000	0.011	0.934	0.019	0.003	0.053
600M	3	0.998	0.020	0.000	0.021	0.964	0.007	0.000	0.017	0.926	0.020	0.003	0.056

Training, validation, and test period error measures for 1MM – 600M PCR models. Input data is not standardized (demeaned and normalized to have unit variance and mean zero). M* per tenor is the optimal number of principal components per validation set $R^2_{Adj.}$. I report R^2 (not adjusted), MAPE, MSE, and RMSE. MAPE over 1 means 100%+. Training period: 2019-05-22 – 2020-12-31 (T=403, 69.84%), Validation period: 2019-02-26 – 2019-05-21 (T=58, 10.05%), Test period: 2018-09-04 – 2019-02-25 (T=116, 20.10%).

Compared to the ABM model, the levels of PCR backfilled 1-, 2-, 3-, and 6-month SOFR are much more plausible for the 2010 – 2014 period. PCA SOFR is below EFFR for the 1,2, and 3-month tenors. For comparability to other models, I do not standardize the panel of input rates when performing PCR. PCR is deterministic. To generate standard errors for the test error measures, as I did for the stochastic ABM and multivariate white noise processes, I would have to nonparametrically resample either the input data or the residuals from PCR. I present a table of test results below for all models, spread and levels.

Model Test Error Comparison

Predict	SOFR – EFFR Spread								SOFR	
Model	ABM		Average		WN		0 (EFFR)		PCR	
Tenor	R^2 (Avg.)	RMSE (Avg.)	R^2	RMSE	R^2 (Avg.)	RMSE (Avg.)	R^2	RMSE	R^2	RMSE
1M	0.583	0.115	0.619	0.110	0.538	0.121	0.610	0.112	0.356	0.144

2M	0.617	0.095	0.648	0.092	0.572	0.101	0.631	0.094	0.389	0.121
3M	0.659	0.076	0.719	0.069	0.627	0.079	0.689	0.073	0.642	0.078
6M	0.647	0.044	0.784	0.035	0.573	0.049	0.685	0.042	0.056	0.073
9M	0.515	0.037	0.756	0.027	0.401	0.042	0.569	0.035	-0.171	0.058
12M	0.723	0.034	0.878	0.023	0.677	0.037	0.761	0.032	0.869	0.023
18M	0.955	0.025	0.975	0.019	0.929	0.032	0.942	0.028	0.847	0.046
24M	0.987	0.018	0.990	0.016	0.970	0.028	0.975	0.026	0.949	0.037
36M	0.994	0.017	0.994	0.017	0.985	0.026	0.987	0.025	0.988	0.024
48M	0.990	0.023	0.992	0.020	0.987	0.027	0.989	0.024	0.982	0.031
60M	0.985	0.028	0.990	0.023	0.985	0.029	0.989	0.025	0.984	0.030
84M	0.976	0.037	0.985	0.029	0.980	0.033	0.988	0.026	0.980	0.033
120M	0.961	0.045	0.974	0.037	0.969	0.040	0.980	0.032	0.992	0.021
180M	0.938	0.054	0.953	0.047	0.948	0.049	0.961	0.043	0.977	0.033
240M	0.924	0.059	0.938	0.053	0.933	0.056	0.946	0.050	0.959	0.043
300M	0.913	0.063	0.926	0.058	0.920	0.060	0.933	0.055	0.949	0.048
360M	0.903	0.065	0.915	0.061	0.909	0.063	0.923	0.058	0.944	0.049
480M	0.891	0.068	0.901	0.065	0.894	0.067	0.909	0.062	0.934	0.053
600M	0.884	0.070	0.892	0.067	0.883	0.070	0.899	0.065	0.926	0.056

Test period error measures for 1MM – 600M arithmetic Brownian motion, average, multivariate normal, no-spread, and PCR models. All but the PCR model predict the SOFR-EFFR spread and add to observed EFFR per tenor to obtained predicted SOFR. PCR predicts SOFR in levels. Test period: 2018-09-04 – 2019-02-25 (T=116, 20.10%). ABM and MVN R^2 and RMSE are averages from varied random draws.

A clear pattern emerges. For 1 month to 60 months, the average spread model has the lowest test error and highest test R^2, although model performance converges towards 60 months. At 24 months, the difference in test R^2s across models is at most 0.025. SOFR-EFFR spreads in the test period have an average and volatility that is decreasing in tenor up to 60 months, meaning larger and more volatile spreads for short tenors. At 60 months, the spread is near zero. The PCR, zero spread, ABM, and white noise models struggle as a result. The constant average spread model performs the best in the test period when the observed spread is most volatile because the alternatives predict less accurate time-varying spreads. The no-spread model is best for 84 months, albeit barely, achieving a 0.003 higher R^2[25] than the average spread model. For 180 months and higher, PCR outperforms all other models. This result can be interpreted as follows: the underlying factor structure has a more stable relationship with the longest SOFR tenors, whereas the volatility in short-term SOFR obscures the relationship and results in poor out of sample performance.

[25] Note, R^2 and MSE (and thus RMSE) are inversely related. R^2 equals 1 – RSS/TSS, where RSS = n*MSE. As MSE increases, RSS increases and R^2 falls. Hence it suffices to compare model R^2 to select a model.

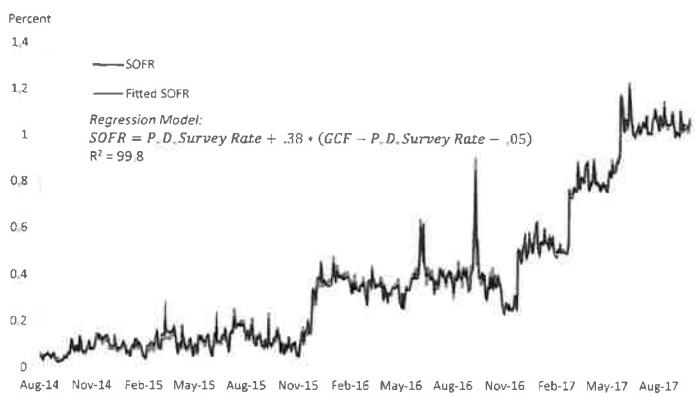
Alternative Data Source

Each model has deficits. Counterintuitively, the average model, which predicts EFFR plus a constant spread equal to the training period observed spread, performs best for tenors with the most volatile spreads. Over long periods, a constant spread between backfilled SOFR and EFFR is not plausible. The ABM and white noise models introduce spread variation but to the detriment of backfilled SOFR, as for most tenors a constant 0 spread model has higher R^2 and lower RMSE. PCR has strong performance for 24 months and higher, and a more compelling explanatory mechanism. However, it performs the worst for short tenors and the coefficients estimated in the regression are, contrary to the assumption, likely time-varying. Each model assumes either EFFR is a valid proxy for SOFR or exhibits strong co-movement. This exercise is motivated by a simple known unknown[26]: what would SOFR be, if it was published before April 2018? The Federal Reserve Bank of New York offers some insight[27] with pre-production SOFR estimates spanning Aug. 2014 – Mar. 2018[28], and a survey rate dating to Feb. 1998 of primary dealers' activity in the Treasury GC repo market. As stated in the FRB note, there are two key differences between the survey rate and SOFR: *"The survey rate is calculated as a volume-weighted mean whereas SOFR is calculated as a volume-weighted median. In addition, the transactions underlying the survey rate are not as broad as those underlying SOFR, as the survey collects only the GC segments of the repo market and does not capture borrowing activity conducted by non-primary dealer market participants"* (Bowman 2019). Unlike EFFR, the survey rate is based on the same economic activity as SOFR. Unlike LIBOR, the survey rate wasn't the reference rate in meaningful financial contracts, so strategic survey responses should not be an issue. The FEDS note models SOFR using the following regression, a function of the FRBNY survey rate and DTCC's GCF rate, which achieves an R^2 of 0.998:

$$\widehat{\text{SOFR}}_t = \text{SurveyRate}_t + 0.38 \cdot (\text{GCF}_t - \text{Survey Rate}_t - 0.05)$$

$$\widehat{\text{SOFR}}_t = 0.62 \cdot \text{SurveyRate}_t + 0.38 \cdot \text{GCF}_t - 0.019$$

Because the survey rate ends in Feb. 2018, and observed SOFR begins in Apr. 2018, I cannot replicate this regression or conduct additional testing. The FEDS note shows a strong in-sample fit:



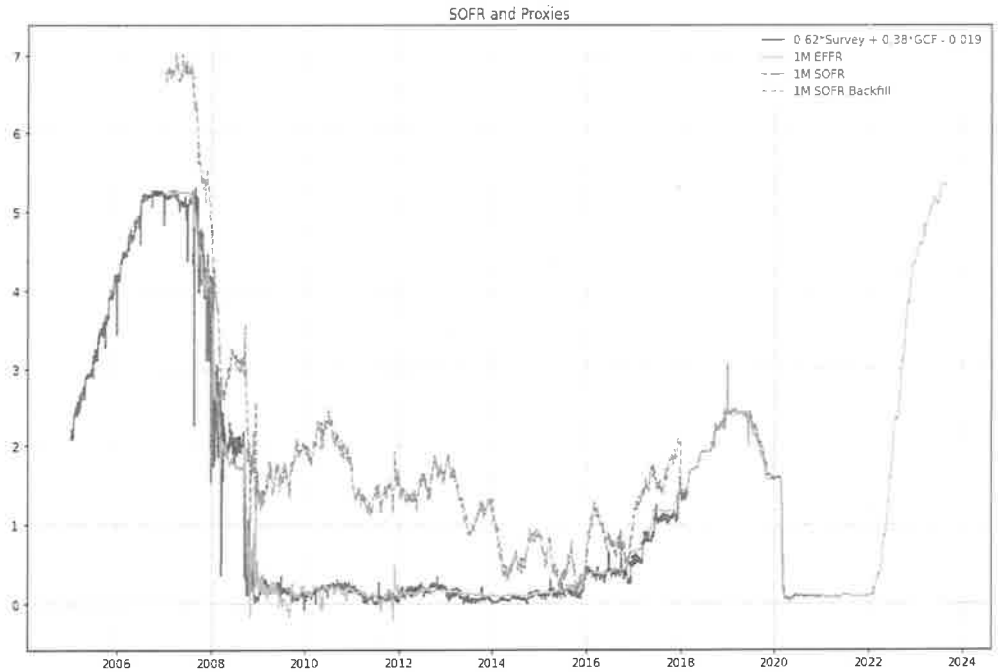
Source: Bowman 2019.

I retrieve the survey rate and DTCC GCF data and use the fitted regression model to backfill 1-month SOFR. Regression fitted SOFR (blue) is close to 1-month EFFR (orange), but shows more volatility and

[26] D. Rumsfeld, Feb. 12th, 2002.

[27] D. Bowman, "Historical Proxies for the Secured Overnight Financing Rate". FEDS Notes, July 15th, 2019. See https://www.federalreserve.gov/econres/notes/feds-notes/historical-proxies-for-the-secured-overnight-financing-rate-20190715.html

[28] Despite seeing several references to this pre-production SOFR data, I couldn't find it.



Time series of 1-month SOFR observed and backfilled, 1-month EFFR, and FEDS Notes fitted regression model.

noticeable decreases during several points in the 2007 – 2009 period. SOFR, had it existed, would likely not have exhibited structural breaks from other reference rates outside of this period. The current backfilled SOFR (grey), using the arithmetic Brownian motion process, is wildly unrealistic in contrast. In 2007, it is nearly 2 percentage points above the fitted rate and EFFR. The large negative changes occur in the survey rate data. Over a three-day window, they are -2.7 percentage points on 12/31/2007, -2.66 percentage points on 12/30/1999, -2.43 percentage points on 3/19/2008, -2.23 percentage points on 8/22/2007, and -2.03 percentage points on 1/23/2008. The end of year decreases shows discrepancies between the survey rate and transaction based SOFR, but the survey rate is nonetheless useful.
In terms of economic interpretation, the survey rate is the closest historical time series to SOFR. I compare each model's backfilled 1-month SOFR to the FRBNY survey rate, the DTCC Treasury GCF Repo Weighted Average Rate, the FEDS Notes regression proxy, and 1-month EFFR as a baseline.

Correlation of Historical Rates with Model Imputed SOFR

ρ	Survey Rate	GCF Rate	Reg. Proxy	EFFR 1M
1M Avg. Spread SOFR	0.990	0.990	0.990	1.000
1M WN Spread SOFR	0.989	0.989	0.989	0.999
1M PCR SOFR	0.986	0.988	0.987	0.997
1M ABM Spread SOFR	0.947	0.943	0.946	0.951
1M No Spread SOFR	0.990	0.990	0.990	1.000

Correlations between observed historical rates and model imputed 1-month SOFR, 1/3/07 – 10/31/16. Survey Rate is FRBNY Survey Rate. GCF is Treasury GCF Repo Weighted Average Rate. 1M No Spread is EFFR. Regression Proxy is the fitted model 0.62*Survey Rate + 0.38*GCF – 0.019.

Constant spread models, either average or 0, have the same correlation of 0.99 with the survey rate. This is a product of the 1-month EFFR's high correlation with the survey rate. The lowest correlation, 0.947, is between the arithmetic Brownian motion model and the survey rate. The ABM SFOR has the lowest correlation with the DTCC GCF Rate, the FEDS Notes regression proxy, and 1-month EFFR.

Per the FEDS note, over Aug. '14 – Aug. '17, coefficients for historical SOFR on the survey and GCF rates are 0.62 and 0.38, respectively, with a constant of -0.019. SOFR can be modelled as a weighted average of the survey and GCF rates, with more weight to the survey rate. If that relationship is correct and time-invariant, then backfilled SOFR from another model should have a similar relationship with the survey and GCF rates, provided backfilled SOFR is accurate. I regress each model's backfilled SOFR on the FRBNY survey rate and the DTCC GCF rate using Jan. '14 – Oct. '16 data[29] and examine the coefficients.

Regression Test: Backfilled 1M SOFR on Survey and GCF Rates, 2014 - 2016

	1M 0 Spread SOFR	1M Avg. Spread SOFR	1M WN Spread SOFR	1M PCR SOFR	1M ABM Spread SOFR
Constant	0.0582*** (0.0061)	0.0820*** (0.0061)	0.0835*** (0.0073)	0.0229 (0.0231)	0.5156*** (0.0390)
Survey Rate	1.4389*** (0.1779)	1.4389*** (0.1779)	1.5693*** (0.1838)	1.6231*** (0.4050)	4.0245*** (0.7176)
Treasury GCF Repo Weighted Average Rate	-0.3671*** (0.1303)	-0.3671*** (0.1303)	-0.4625*** (0.1308)	-0.1914 (0.3104)	-2.2539*** (0.4911)
R^2	0.9002	0.9002	0.6923	0.7269	0.2714

Coefficient estimates for $\widehat{SOFR}_t = \beta_0 + \beta_1 SurveyRate_t + \beta_2 GCF_t + \varepsilon_t$. T=709, 01/02/14 – 10/31/16. Standard errors in parenthesis, stars indicate significance: ***: 1%, **, 5%, *:10%. Newey-West standard errors, $T^{0.25}$ lags.

Over a similar period, each model's backfilled SOFR has a different relationship[30] with the survey rate and GCF repo rate. Each backfilled SOFR series has an inverse relationship with the GCF rate, except for the PCR model, for which the GCF coefficient estimate is statistically indistinguishable from zero. Survey rate coefficients are all significant at 1% level and over 1. The EFFR plus spread models achieve the highest R^2 of 0.9002. This regression does not reject the plausibility of any backfilled SOFR series except the arithmetic Brownian motion. *These results suggest the use of the FEDS proxy rate itself for 1 month.*

[29] See the Appendix for a full sample regression test. PCR SOFR has an inverse relation with the survey rate and direct with the GCF rate, but the constant and white noise spread model' SOFR has no statistically significant relationship with the survey rate. Consistent with the FEDS Notes regression, each model except the ABM achieves an R^2 over 0.97.

[30] The no spread and average spread models have the same coefficients and R^2 but different constant estimates. Both models predict 1M EFFR plus a constant spread, so one is the other vertically displaced. Consequently, the difference in the constant estimates, 0.0820 – 0.0582, is exactly the value of the average SOFR-EFFR spread, 0.0237, used to construct backfilled SOFR in the average spread model.

Discussion

I outline several key issues.

First, backfilling is a counterfactual exercise, as it asks what SOFR would be if it had been computed and published well before it was. Predicting future values of a macroeconomic time series mildly assumes that the future bears tangible resemblance to the observed past. In contrast, whether the past resembles current times is an empirical question. Backfilling is complicated because the imputed values should cohere with the stylized facts of the known past. If reference rates exhibited a structural break during the financial crisis, so should SOFR. The level of SOFR during this period should be close to that of comparable reference rates; backfilled SOFR several hundred basis points above the historical effective federal funds rate would be wildly implausible. There is no guarantee that the relationship between SOFR and contemporaneous covariates, such as other reference rates, from 2018 forward would have held before 2018. Had SOFR existed a decade before its introduction, the relationship it has with other reference rates may have manifested differently over time, evident in SOFR's increased trading volume. The challenges associated with this exercise tempers expectations for backfilled SOFR's quality.

Second, evaluating the quality of a backfilled time series is another complication. SOFR doesn't exist prior to 2018, so there is no true value to compare backfilled SOFR to. Using the backfilling methodology to fill forward from 2018 is insufficient, because the methodology is based upon the same period, effectively constituting in sample testing. SOFR has less than 6 years of historical data, hence a temporal approach where in the testing and validation periods precede the training period still leaves a much longer out of sample prediction period. Comparing functions of the imputed series, like VaR, is akin to comparing reflections. It begs the question of whether those external tests are sensitive enough to detect aberrational imputed values. The likely are not. It is difficult to discern whether an external test of a forward filled and historical series shows no difference because the test is simply insensitive to any difference that exists, possibly of great magnitude, or because there is no difference. Establishing a temporal testing, validation, and training set using the 2018 forward observed data has its flaws, namely the unusual volatility of SOFR in the early testing period, but is nonetheless the only direct test of a model's backfilling ability. Alternatives, such as a randomized split, are worse or irrelevant.

Third, spread models, which add a predicted spread to a proxy rate, have a distinct advantage to the extent that the proxy rate is suitable. The proxy rate preserves the appropriate level and co-movement, both cross-tenor and with the target rate, for the backfilled rate, but shoddy spreads can completely distort both. A dynamic spread like that of the ABM process is evidence of this very issue. The utility of a time-varying spread between backfilled SOFR and another reference rate, such as EFFR, should be weighed against the lessened quality of backfilled SOFR. Poor spreads, constant or dynamic, do not necessarily exculpate proxy rates. Another upside of this paper's testing methodology is that it reveals some error in SOFR backfilled with spread models is attributable to the proxy rate itself.

Fourth, among models that backfill SOFR directly, there is no shortage of specifications and potential covariates. Predicting forwards or backwards a continuous series is a regression problem, for which numerous models are applicable. I favor principal components regression because of its prevalence in modelling the term structure and for the desirable properties of ordinary least squares. I disfavor time series models because they exhibit poor in-sample fit[31] and perform poorly over long backfilling periods.

[31] More complicated time-series models for the SOFR – EFFR spread are not viable. For the 9/4/18 – 12/30/20 period, spreads in levels are compressed between a 0.10% and 0.20% range and exhibit time-varying volatility. Short tenors (1M, 2M, 3M) and long tenors (300M, 360M, 480M) are stationary, but the remaining tenors are non-stationary, so they must be first-differenced to model with an ARMA(p,q) process. The first differenced spreads however are nearly 0 and resemble white noise, making them quite difficult to predict. Spreads show serial

Fifth, backfilling processes that involve the cumulative product or sum of random draws exhibit ensemble variance increasing in time. It is imperative to distinguish between the time-varying volatility of a single backfilled series, evident in rolling volatility estimates, and the dispersion in values from multiple simulated paths at the same period. Repeatedly using the ABM process can result in wildly different backfilled SOFR paths for the same tenor, a product exclusively of random variation. After subtracting the proxy rate, the paths bear little resemblance to each other. Moreover, the economic interpretation of such backfilled series is that their ensemble variance mechanically decreases as time progresses, a trait usually not evident in macroeconomic time series. ABM SOFR shows decreasing ensemble variance from 2007 to 2018, a spike in actual volatility around its introduction, and then reduced but still time-varying volatility. Deterministic spread models show no ensemble variance. Deterministic spread models have near zero computational overheard and are perfectly interpretable.

Sixth, what SOFR would've been had the index been published beginning in 2007 constitutes a known unknown. Whether or not SOFR, had it existed, exhibits a structural break with its proxy rate constitutes an unknown unknown. The FRBNY survey rate provides the closest to an answer for both difficult questions. The survey rate has the most similar economic interpretation to SOFR, compared to other proxy rates, and the regression model – a linear combination of the survey rate and the DTCC GFC rate - showed in the FRBNY FEDS Notes post has strong in sample fit. Both show a similar level to EFFR and no structural breaks. Despite the slight differences with SOFR, the survey rate and regression model are documented and referenced on the FRBNY site and do not introduce comparable model risk.

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correlation primarily through the first lag and mild evidence of an ARCH effect, based upon the ACF of squared residuals. Spreads are not normally distributed. The 9-month to 300-month distributions resemble beta distributions with positive skew, undermining the case for the multivariate normal draw model. See the Appendix for time-series properties of SOFR – EFFR spreads.

Conclusion

Backfilling SOFR is one component of a broader exercise to populate the historical time series of alternative reference rates during the period prior to their publication.
current approach and methodology, " (2023), backfills SOFR by adding a stochastic spread to the effective federal funds rate. The quality of backfilled SOFR bears directly on volatility and correlation calculations used to generate VaR and SVaR figures. For all analysis, I use reference rate data, including the currently used backfilled SOFR series, from and provided by . Despite preserving cross-tenor correlation, the volatility and levels of the current backfilled SOFR-EFFR spreads are wildly different from observed SOFR-EFFR spreads at all tenors. Even if implemented correctly, the arithmetic Brownian motion process produces a SOFR-EFFR spread that exhibits increasing variance. Repeated draws from the process result in wildly different spreads and thus backfilled SOFR series per tenor. The spreads are large and volatile enough to obscure the otherwise close relationship between SOFR and EFFR at the same tenor, and for medium and long-tenors this results in a worse backfilled SOFR series than just using EFFR.

Each assessment of the quality of backfilled SOFR takes the form of comparing properties of observed SOFR, such as cross-tenor correlation, correlation with EFFR, summary statistics, effect on VaR, with forward filled SOFR or backward filled SOFR. This poses two problems: first, it effectively constitutes in sample testing, and second, it begs the question of whether the aforementioned comparisons are sensitive enough to detect inaccurate forward or backward filled SOFR. I use a new process to evaluate several backfilling methods, by segmenting the 9/4/18 - 12/31/20 period into disjoint test (9/4/18 - 2/25/19) and training periods (2/26/19-12/31/20), with a temporal validation set carved out from the training period if needed. I train models on the most recent observed SOFR data, predict over the test period, and finally predict over the long out-of-sample pre-2018 period. This is the only way to obtain concrete test error measures for SOFR itself. I evaluate five models. Four models (arithmetic Brownian motion, white noise, average spread, zero spread) backfill SOFR by predicting a SOFR-EFFR spread and then adding EFFR, the fifth (principal components regression) backfills SOFR itself. Stochastic spread models - the arithmetic Brownian motion and white noise models - perform abysmally at predicting spreads, as it to be expected, but achieve respectable performance in the test set for SOFR because of the proximity in levels and co-movement between SOFR and EFFR at the same tenor. Deterministic spread models - a constant spread equal to the training period average or zero - outperform random spread models, and the average spread model outperforms the zero-spread model at short tenors but not at long tenors. The tradeoff with using a stochastic spread model simply to induce time-varying spreads between reference rates is a noisier backfilled SOFR series. Backfilled SOFR shouldn't be the corresponding EFFR, because there is empirically a significant spread and mechanically it would pose issues for risk systems to have a constant zero spread. Principal components regression achieves the best performance for long tenors, as it identifies a stable relationship between SOFR and the underlying factor structure. I recommend using an average spread model to backfill SOFR for the 2M - 60M tenors and principal components regression for the 120M - 600M tenors.

What would SOFR be before it was published before 2018 is a question that can only be answered probabilistically. However, the Federal Reserve Bank of New York released a historical survey rate of primary dealers' activity in the Treasury GC repo market which provides the most definitive answer. This survey rate has the same economic interpretation as SOFR, and a post on the FRBNY site shows a regression model - a combination of the survey rate and the DTCC GCF repo rate - that closely mimics pre-production SOFR from 8/2014 - 08/2017. This fitted model should be used to backfill 1-month SOFR.

Appendix

Fat-Tail Scaling

The diffusion parameter in an arithmetic Brownian motion process requires calibration. The MDD calibrates σ^{τ}, the per tenor diffusion, using observed SOFR – EFFR spread data in the period 9/4/18 – 1/31/21. To account for more mass in the tails of the distribution of the spread change, the MDD adopts fat-tail scaling, which takes the maximum of the absolute standardized 99th percentile, absolute standardized 1st percentile, and volatility of the daily *change* in spread as follows[32].

$$\sigma_{FST} = \max\left[\sigma_{\Delta S}, \frac{|\Delta S_1 - \overline{\Delta S}|}{2.326}, \frac{|\Delta S_{99} - \overline{\Delta S}|}{2.326}\right]$$

I use this methodology and track the actual volatility measure used. Left tail volatility is highest for most medium tenors, ranging from 12 months to 180 months, except for 24 months. Right tail volatility is highest for the long tenors 300 months, 360 months, and 600 months. The sample standard deviation is highest for the short tenors, 2 months to 9 months. Fat-tail scaled volatility can be much higher than the

SOFR-EFFR Spread Change Volatility Measures

Tenor	Std. Dev.	Left Tail	Right Tail	FTS Vol.	FTS Vol./Vol
1M	0.040518	0.041378	0.038391	Left Tail	1.021211
2M	0.034308	0.029461	0.030585	Std. Dev.	1.000000
3M	0.028415	0.020875	0.024247	Std. Dev.	1.000000
6M	0.016359	0.013138	0.013614	Std. Dev.	1.000000
9M	0.011422	0.011154	0.009937	Std. Dev.	1.000000
12M	0.008850	0.009851	0.007725	Left Tail	1.113167
18M	0.006281	0.007653	0.006987	Left Tail	1.218512
24M	0.005090	0.007001	0.007662	Right Tail	1.505468
36M	0.003953	0.005543	0.005343	Left Tail	1.402157
48M	0.003356	0.004882	0.004349	Left Tail	1.454704
60M	0.003039	0.004218	0.003956	Left Tail	1.387844
84M	0.002799	0.004050	0.003424	Left Tail	1.446694
120M	0.003021	0.004456	0.003414	Left Tail	1.475078
180M	0.003672	0.003911	0.003761	Left Tail	1.065055
240M	0.004110	0.003696	0.003874	Std. Dev.	1.000000
300M	0.004365	0.003640	0.004432	Right Tail	1.015305
360M	0.004556	0.003948	0.004639	Right Tail	1.018037
480M	0.004806	0.004100	0.004484	Std. Dev.	1.000000
600M	0.005058	0.004420	0.005289	Right Tail	1.045677

Volatilities used in fat-tail scaling for calibration of spread arithmetic Brownian motion diffusion parameter. Computed using 9/4/18 – 12/31/20 daily change in SOFR-EFFR spread data. Left tail is absolute standardized 1st percentile spread changes, right tail is absolute standardized 99th percentile spread changes. Standard deviation is sample standard deviation. FTS volatility is the largest of the three.

sample standard deviation, up to 50% higher and around 40% higher for medium tenors. Fat-tail scaling is a conservative approach to diffusion parameter calibration, but the downside for some tenors is considerably higher volatility in the backfilled spreads than warranted by observed spreads. I compute the sample skewness and kurtosis for spread changes over the same period. A normal distribution has 0 skewness and kurtosis of 3 (mesokurtic, with excess kurtosis 0). In contrast, the spread change

[32] 2.326 and -2.326 are the 99% and 1% critical values for a Z-distribution, P(Z <= -2.326) = 0.01, P(Z >= 2.326) = 0.01 for Z~N(0,1)

distributions are leptokurtic, exhibiting high positive excess kurtosis. The 84-month tenor exhibits the lowest excess kurtosis of 10.8. All tenors exhibit skewness; the 1-month, 2-month, and 3-month tenors have positive skewness, and the remaining tenors have negative skewness.

SOFR-EFFR Spread Change Skewness and Kurtosis

Tenor	1M	2M	3M	6M	9M	12M	18M	24M	36M
Skewness	2.786	3.226	3.510	-1.194	-1.139	-1.096	-0.923	-0.715	-0.584
Kurtosis	71.536	82.452	85.510	90.350	79.756	70.699	55.157	42.882	26.342

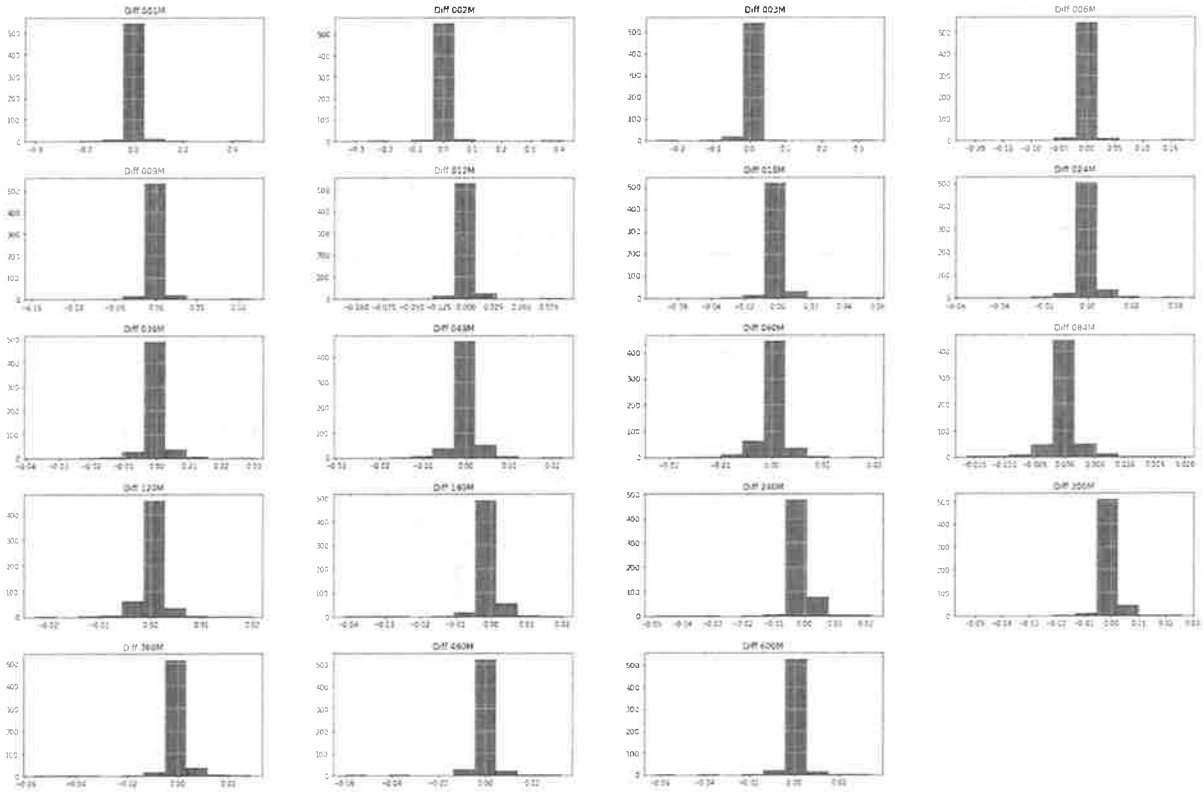
Sample skewness and kurtosis, daily change in SOFR-EFFR spreads 9/4/18 – 12/30/21. 1M-36M tenors.

Tenor	48M	60M	84M	120M	180M	240M	300M	360M	480M	600M
Skewness	-0.511	-0.294	-0.048	-1.083	-3.797	-4.753	-5.218	-5.388	-5.491	-5.204
Kurtosis	19.842	16.374	13.765	18.503	46.862	59.476	67.172	70.918	74.891	71.315

Sample skewness and kurtosis, daily change in SOFR-EFFR spreads 9/4/18 – 12/30/21. 48M-600M tenors.

Skewness in magnitude is largest for the shortest and longest tenors. The negative skewness of medium tenors, 36 months to 180 months, matches the use of left tail volatility. I conclude that the use of fat-tail scaling is warranted. It either makes little difference to the choice of diffusion parameter estimate (1M – 9M, 240M – 600M tenors) or coheres with the standardized sample moments (36M – 180M).

Histograms: Change in SOFR-EFFR Spread Per Tenor



Distributions of the daily change in SOFR-EFFR spreads per tenor, 9/4/18 – 12/31/20. T=576. 1M – 600M tenors.

Zero Backfilled SOFR-EFFR Spreads

SOFR – EFFR Spread Per Tenor, 12/26/2017 – 01/09/2018

Date	12/26/17	12/27/17	12/28/17	12/29/17	1/2/18	1/3/18	1/4/18	1/5/18	1/8/18	1/9/18
001M	0.647	0.593	0.563	0.557	0.000	0.000	0.000	0.000	0.000	0.000
002M	0.619	0.562	0.529	0.532	-0.000	-0.000	-0.000	-0.000	-0.000	0.000
003M	0.534	0.479	0.450	0.456	0.000	-0.000	-0.000	-0.000	-0.000	0.000
006M	0.382	0.359	0.349	0.363	-0.000	0.000	0.000	-0.000	-0.000	0.000
009M	0.240	0.225	0.219	0.228	-0.000	-0.000	0.000	0.000	0.000	0.000
012M	0.199	0.186	0.181	0.189	-0.000	-0.000	-0.000	0.000	-0.000	0.000
018M	0.154	0.146	0.143	0.150	-0.000	0.000	-0.000	0.000	-0.000	-0.000
024M	0.123	0.114	0.112	0.119	0.000	0.000	0.000	0.000	0.000	-0.000
036M	0.061	0.056	0.055	0.058	0.000	-0.000	-0.000	-0.000	-0.000	0.000
048M	0.037	0.033	0.032	0.033	0.000	-0.000	-0.000	0.000	0.000	-0.000
060M	0.024	0.020	0.019	0.019	0.000	-0.000	-0.000	-0.000	-0.000	0.000
084M	0.044	0.039	0.037	0.035	0.000	-0.000	-0.000	0.000	0.000	0.000
120M	0.072	0.067	0.066	0.062	-0.000	-0.000	0.000	-0.000	0.000	0.000
180M	0.072	0.070	0.069	0.067	0.000	0.000	0.000	0.000	0.000	-0.000
240M	0.070	0.069	0.068	0.067	-0.000	-0.000	0.000	-0.000	0.000	0.000
300M	0.076	0.075	0.074	0.073	-0.000	0.000	-0.000	-0.000	0.000	-0.000
360M	0.083	0.082	0.082	0.080	0.000	0.000	0.000	-0.000	0.000	-0.000
480M	0.098	0.096	0.095	0.093	0.000	-0.000	0.000	-0.000	-0.000	-0.000

SOFR – EFFR Spread Per Tenor, 12/26/2017 – 1/9/2018

Date	8/20/18	8/21/18	8/22/18	8/23/18	8/24/18	8/27/18	8/28/18	8/30/18	8/31/18	9/4/18
001M	0.000	0.000	0.000	-0.000	0.000	0.000	0.095	0.042	0.041	0.044
002M	0.000	-0.000	-0.000	0.000	0.000	-0.000	0.020	-0.000	0.000	0.009
003M	-0.000	-0.000	0.000	0.000	0.000	-0.000	0.019	0.011	0.011	0.017
006M	0.000	0.000	-0.000	0.000	0.000	0.000	0.016	0.013	0.013	0.016
009M	-0.000	0.000	0.000	0.000	0.000	0.000	0.018	0.016	0.015	0.017
012M	0.000	0.000	0.000	-0.000	0.000	0.000	0.018	0.016	0.016	0.018
018M	-0.000	-0.000	-0.000	-0.000	-0.000	0.000	0.018	0.017	0.016	0.017
024M	-0.000	-0.000	-0.000	0.000	0.000	-0.000	0.017	0.016	0.016	0.017
036M	-0.000	-0.000	-0.000	-0.000	-0.000	0.000	0.015	0.014	0.014	0.015
048M	-0.000	-0.000	-0.000	-0.000	-0.000	0.000	0.014	0.013	0.013	0.014
060M	-0.000	-0.000	0.000	-0.000	0.000	-0.000	0.013	0.013	0.013	0.013
084M	0.000	-0.000	-0.000	0.000	0.000	-0.000	0.012	0.012	0.012	0.012
120M	0.000	0.000	0.000	0.000	-0.000	0.000	0.012	0.011	0.011	0.012
180M	0.000	0.000	-0.000	0.000	-0.000	0.000	0.011	0.011	0.011	0.011
240M	-0.000	-0.000	0.000	-0.000	-0.000	-0.000	0.011	0.011	0.011	0.011
300M	0.000	0.000	0.000	-0.000	-0.000	-0.000	0.011	0.011	0.011	0.011
360M	0.000	-0.000	0.000	0.000	-0.000	0.000	0.011	0.011	0.011	0.011
480M	0.000	0.000	0.000	-0.000	-0.000	-0.000	0.011	0.010	0.010	0.011

SOFR-EFFR Spread Summary Statistics

01/03/2007 – 8/31/2018 1M – 36M Backfilled Spread Summary Statistics

Tenor	1M	2M	3M	6M	9M	12M	18M	24M	36M
T	2,935	2,935	2,935	2,935	2,935	2,935	2,929	2,935	2,935
Mean	0.991	0.725	0.527	0.245	0.072	0.000	-0.053	-0.152	-0.173
Std.	0.582	0.472	0.406	0.177	0.106	0.098	0.088	0.114	0.090
Min.	-0.132	-0.297	-0.387	-0.132	-0.142	-0.210	-0.253	-0.399	-0.333
P1	-0.005	-0.188	-0.312	-0.090	-0.113	-0.189	-0.227	-0.367	-0.314
P25	0.485	0.340	0.196	0.123	-0.008	-0.075	-0.111	-0.221	-0.234
P50	1.127	0.809	0.576	0.242	0.057	-0.000	-0.059	-0.169	-0.197
P75	1.439	1.070	0.832	0.357	0.151	0.066	0.000	-0.082	-0.135
P99	2.124	1.703	1.339	0.679	0.323	0.224	0.170	0.134	0.071
Max.	2.315	1.835	1.455	0.754	0.376	0.251	0.195	0.161	0.090

01/03/2007 – 8/31/2018 48M – 480M Backfilled Spread Summary Statistics (T=2,395)

Tenor	48	60	84	120	180	240	300	360	480M
Mean	-0.220	-0.236	-0.183	-0.117	-0.011	0.040	0.075	0.102	0.108
Std.	0.104	0.111	0.097	0.085	0.055	0.060	0.066	0.071	0.074
Min.	-0.390	-0.410	-0.353	-0.289	-0.158	-0.123	-0.097	-0.080	-0.059
P1	-0.374	-0.397	-0.342	-0.278	-0.130	-0.096	-0.071	-0.052	-0.030
P25	-0.288	-0.307	-0.249	-0.180	-0.060	-0.004	0.018	0.044	0.043
P50	-0.245	-0.261	-0.191	-0.104	-0.000	0.040	0.079	0.107	0.112
P75	-0.187	-0.206	-0.147	-0.061	0.040	0.097	0.136	0.165	0.178
P99	0.043	0.027	0.042	0.067	0.084	0.137	0.181	0.211	0.233
Max.	0.064	0.050	0.066	0.092	0.097	0.154	0.197	0.220	0.243

09/04/2018 – 12/31/2020 1M – 36M Historical Spread Summary Statistics (T=577)

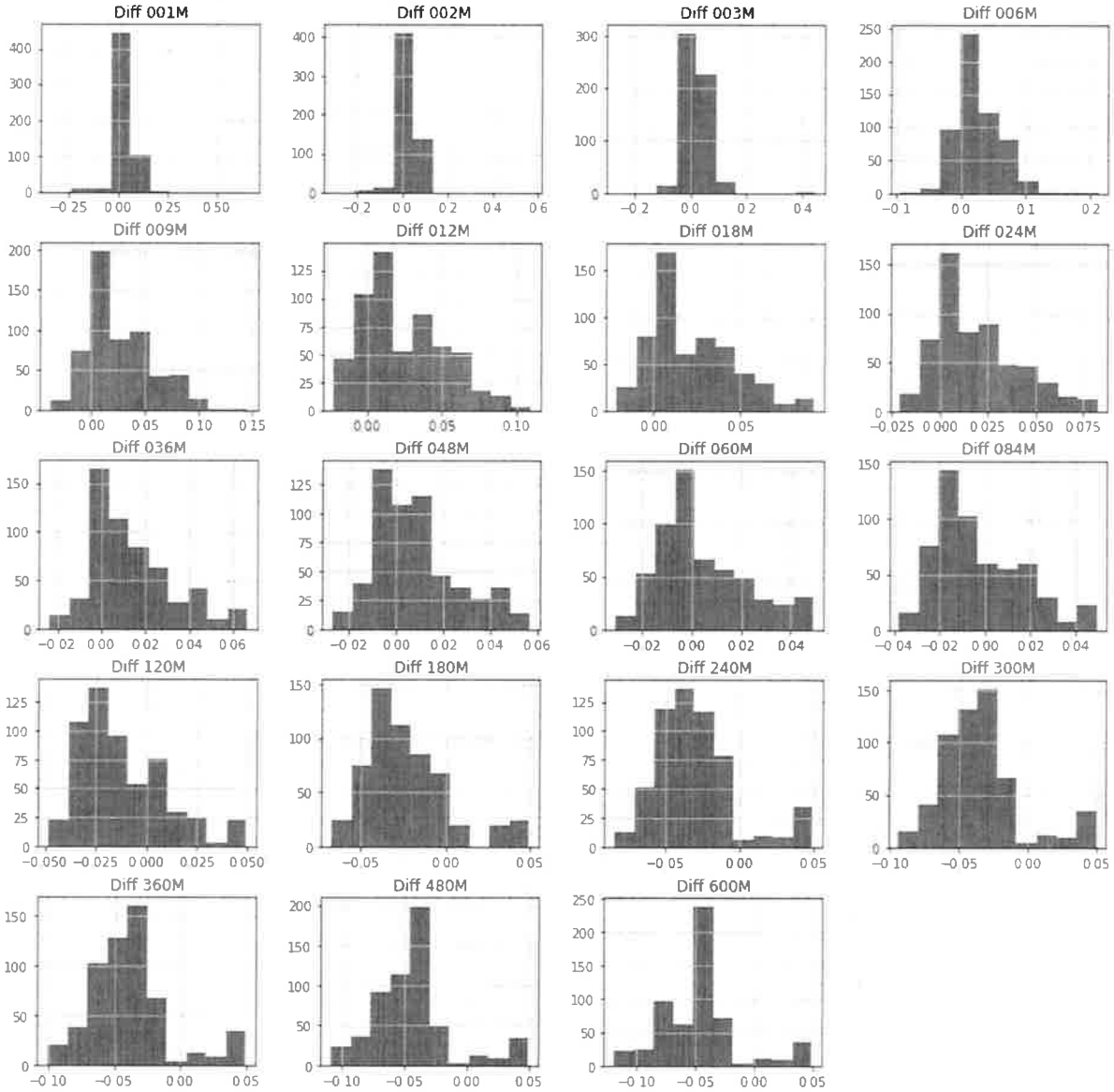
Tenor	1M	2M	3M	6M	9M	12M	18M	24M	36M
Mean	0.024	0.023	0.025	0.026	0.026	0.024	0.022	0.019	0.014
Std.	0.067	0.056	0.047	0.035	0.031	0.028	0.024	0.022	0.019
Min.	-0.340	-0.303	-0.264	-0.093	-0.038	-0.023	-0.021	-0.021	-0.024
P1	-0.189	-0.134	-0.090	-0.035	-0.022	-0.021	-0.021	-0.021	-0.021
P25	-0.006	-0.003	-0.001	0.003	0.003	0.003	0.003	0.002	-0.002
P50	0.011	0.013	0.017	0.018	0.017	0.016	0.015	0.014	0.011
P75	0.050	0.047	0.049	0.048	0.045	0.043	0.039	0.033	0.025
P99	0.169	0.122	0.113	0.111	0.104	0.096	0.086	0.076	0.062
Max.	0.665	0.575	0.448	0.212	0.147	0.110	0.093	0.083	0.067

09/04/2018 – 12/31/2020 48M – 480M Historical Spread Summary Statistics (T=577)

Tenor	48M	60M	84M	120M	180M	140M	300M	360M	480M
Mean	0.009	0.005	-0.003	-0.012	-0.023	-0.030	-0.034	-0.038	-0.043
Std.	0.018	0.018	0.019	0.022	0.025	0.027	0.029	0.031	0.033
Min.	-0.027	-0.031	-0.038	-0.048	-0.068	-0.084	-0.094	-0.101	-0.109
P1	-0.023	-0.026	-0.033	-0.044	-0.064	-0.079	-0.090	-0.097	-0.106
P25	-0.004	-0.008	-0.017	-0.028	-0.041	-0.048	-0.054	-0.058	-0.065
P50	0.005	-0.000	-0.007	-0.018	-0.029	-0.033	-0.038	-0.041	-0.044
P75	0.019	0.016	0.012	0.004	-0.010	-0.017	-0.024	-0.028	-0.032
P99	0.052	0.049	0.049	0.049	0.049	0.050	0.050	0.050	0.050
Max.	0.057	0.049	0.049	0.050	0.050	0.050	0.050	0.050	0.050

Observed SOFR-EFFR Spread Distributions

Histograms: SOFR-EFFR Spread Distributions per Tenor, 9/4/18 – 12/31/20



SOFR-EFFR Spread Cross-Tenor Correlation

SOFR-EFFR Spread Correlation, 1M-48M

	1M	2M	3M	6M	9M	12M	18M	24M	36M	48M
1M	1.000	0.976	0.942	0.805	0.718	0.677	0.643	0.622	0.574	0.526
2M	0.976	1.000	0.980	0.862	0.785	0.747	0.714	0.693	0.644	0.589
3M	0.942	0.980	1.000	0.924	0.863	0.827	0.791	0.764	0.701	0.629
6M	0.805	0.862	0.924	1.000	0.986	0.969	0.943	0.916	0.842	0.744
9M	0.718	0.785	0.863	0.986	1.000	0.995	0.975	0.949	0.871	0.765
12M	0.677	0.747	0.827	0.969	0.995	1.000	0.990	0.968	0.895	0.791
18M	0.643	0.714	0.791	0.943	0.975	0.990	1.000	0.991	0.938	0.848
24M	0.622	0.693	0.764	0.916	0.949	0.968	0.991	1.000	0.973	0.905
36M	0.574	0.644	0.701	0.842	0.871	0.895	0.938	0.973	1.000	0.977
48M	0.526	0.589	0.629	0.744	0.765	0.791	0.848	0.905	0.977	1.000
60M	0.469	0.525	0.550	0.642	0.656	0.683	0.750	0.821	0.925	0.985
84M	0.374	0.417	0.427	0.491	0.499	0.528	0.603	0.686	0.823	0.922
120M	0.276	0.311	0.317	0.373	0.386	0.420	0.496	0.583	0.728	0.843
180M	0.178	0.208	0.218	0.278	0.302	0.340	0.412	0.494	0.635	0.750
240M	0.129	0.160	0.172	0.236	0.267	0.306	0.374	0.452	0.587	0.699
300M	0.097	0.128	0.142	0.206	0.241	0.281	0.346	0.420	0.552	0.662
360M	0.078	0.109	0.124	0.189	0.227	0.267	0.331	0.403	0.532	0.640
480M	0.055	0.087	0.103	0.169	0.210	0.250	0.311	0.381	0.506	0.610
600M	0.043	0.075	0.092	0.159	0.201	0.242	0.301	0.369	0.491	0.594

Cross-tenor correlations for SOFR-EFFR spread, 1M-48M. Computed on 9/4/18-12/31/20 data. Column max bolded.

SOFR-EFFR Spread Correlation, 60M-600M

	60M	84M	120M	180M	240M	300M	360M	480M	600M
1M	0.469	0.374	0.276	0.178	0.129	0.097	0.078	0.055	0.043
2M	0.525	0.417	0.311	0.208	0.160	0.128	0.109	0.087	0.075
3M	0.550	0.427	0.317	0.218	0.172	0.142	0.124	0.103	0.092
6M	0.642	0.491	0.373	0.278	0.236	0.206	0.189	0.169	0.159
9M	0.656	0.499	0.386	0.302	0.267	0.241	0.227	0.210	0.201
12M	0.683	0.528	0.420	0.340	0.306	0.281	0.267	0.250	0.242
18M	0.750	0.603	0.496	0.412	0.374	0.346	0.331	0.311	0.301
24M	0.821	0.686	0.583	0.494	0.452	0.420	0.403	0.381	0.369
36M	0.925	0.823	0.728	0.635	0.587	0.552	0.532	0.506	0.491
48M	0.985	0.922	0.843	0.750	0.699	0.662	0.640	0.610	0.594
60M	1.000	0.974	0.914	0.826	0.775	0.738	0.714	0.684	0.666
84M	0.974	1.000	0.976	0.910	0.865	0.830	0.807	0.776	0.758
120M	0.914	0.976	1.000	0.975	0.945	0.919	0.900	0.873	0.857
180M	0.826	0.910	0.975	1.000	0.993	0.981	0.970	0.952	0.940
240M	0.775	0.865	0.945	0.993	1.000	0.997	0.991	0.980	0.971
300M	0.738	0.830	0.919	0.981	0.997	1.000	0.998	0.992	0.986
360M	0.714	0.807	0.900	0.970	0.991	0.998	1.000	0.997	0.993
480M	0.684	0.776	0.873	0.952	0.980	0.992	0.997	1.000	0.999
600M	0.666	0.758	0.857	0.940	0.971	0.986	0.993	0.999	1.000

Cross-tenor correlations for SOFR-EFFR spread, 60M-600M. Computed on 9/4/18-12/31/20 data. Column max bolded.

Correlation Matrix Difference SOFR EFFR Historical and Backfilled

9/4/18 – 12/30/21 SOFR Correlation with EFFR 1/03/07 - 9/4/18 SOFR Correlation with EFFR, 1M-48M

	E-1M	E-2M	E-3M	E-6M	E-9M	E-12M	E-18M	E-24M	E-36M	E-48M
S-1M	0.070	0.069	0.068	0.065	0.060	0.054	0.044	0.039	0.040	0.043
S-2M	0.050	0.049	0.047	0.046	0.042	0.036	0.026	0.022	0.027	0.037
S-3M	0.036	0.035	0.034	0.034	0.030	0.024	0.016	0.013	0.021	0.034
S-6M	0.005	0.006	0.007	0.008	0.005	0.001	-0.004	-0.003	0.016	0.043
S-9M	-0.005	-0.002	0.000	0.003	0.003	0.001	-0.002	0.002	0.026	0.058
S-12M	-0.010	-0.006	-0.003	0.002	0.003	0.002	0.001	0.006	0.031	0.065
S-18M	-0.017	-0.012	-0.008	0.001	0.003	0.003	0.002	0.004	0.025	0.057
S-24M	-0.014	-0.009	-0.005	0.005	0.009	0.008	0.004	0.003	0.018	0.045
S-36M	0.000	0.006	0.010	0.024	0.028	0.027	0.017	0.008	0.002	0.014
S-48M	0.023	0.028	0.033	0.049	0.054	0.053	0.039	0.024	0.004	0.003

9/4/18 – 12/30/21 SOFR Correlation with EFFR 1/03/07 - 9/4/18 SOFR Correlation with EFFR, 60M-600M

	E-60M	E-84M	E-120M	E-180M	E-240M	E-300M	E-360M	E-480M	E-600M
S-60M	0.004	0.027	0.072	0.131	0.158	0.175	0.187	0.196	0.196
S-84M	0.010	0.003	0.020	0.056	0.075	0.088	0.096	0.103	0.103
S-120M	0.049	0.013	0.002	0.012	0.022	0.029	0.034	0.038	0.038
S-180M	0.111	0.051	0.014	0.001	0.001	0.003	0.004	0.006	0.006
S-240M	0.142	0.074	0.027	0.005	0.001	0.000	0.001	0.001	0.001
S-300M	0.162	0.088	0.037	0.009	0.003	0.001	0.001	0.000	0.000
S-360M	0.173	0.097	0.043	0.012	0.005	0.003	0.001	0.001	0.001
S-480M	0.185	0.106	0.049	0.015	0.007	0.004	0.002	0.002	0.002
S-600M	0.185	0.106	0.049	0.015	0.007	0.004	0.002	0.001	0.002

SOFR Training, Testing, Validation Period Summary Statistics

Training, Validation, Testing Period SOFR Summary Statistics: 1M – 48M

Tenor	1M	2M	3M	6M	9M	12M	18M	24M	36M	48M
Median										
Test	2.286	2.327	2.359	2.440	2.473	2.498	2.569	2.618	2.641	2.645
Train	0.244	0.209	0.192	0.161	0.150	0.147	0.172	0.201	0.286	0.362
Validation	2.431	2.431	2.428	2.414	2.386	2.354	2.272	2.206	2.125	2.096
Maximum										
Test	3.064	2.969	2.838	2.606	2.569	2.660	2.766	2.833	2.890	2.915
Train	2.450	2.443	2.428	2.378	2.330	2.275	2.163	2.082	1.997	1.968
Validation	2.473	2.455	2.461	2.469	2.485	2.495	2.459	2.426	2.365	2.339
Mean										
Test	2.299	2.323	2.346	2.412	2.472	2.521	2.561	2.580	2.578	2.573
Train	0.971	0.946	0.927	0.876	0.836	0.802	0.750	0.719	0.701	0.716
Validation	2.428	2.428	2.425	2.411	2.387	2.361	2.284	2.221	2.143	2.114
Minimum										
Test	1.971	2.032	2.091	2.233	2.329	2.410	2.353	2.277	2.184	2.153
Train	0.013	0.018	0.022	0.025	0.025	0.012	-0.006	-0.018	-0.022	-0.010
Validation	2.379	2.377	2.378	2.337	2.281	2.227	2.108	2.023	1.940	1.916
Standard Deviation										
Test	0.180	0.155	0.131	0.076	0.054	0.065	0.119	0.165	0.215	0.232
Train	0.944	0.918	0.899	0.850	0.813	0.781	0.733	0.704	0.669	0.640
Validation	0.018	0.019	0.019	0.031	0.050	0.068	0.091	0.105	0.110	0.108

Summary statistics for SOFR 1M – 48M per sample period. Test sample is 2018-09-04 to 2019-02-25 (T=116, 20.10%), validation sample is 2019-02-26 to 2019-05-21 (T=58, 10.05%), train sample is 2019-05-22 – 2020-12-31 (T=403, 69.84%). All tenors non-stationary for all samples.

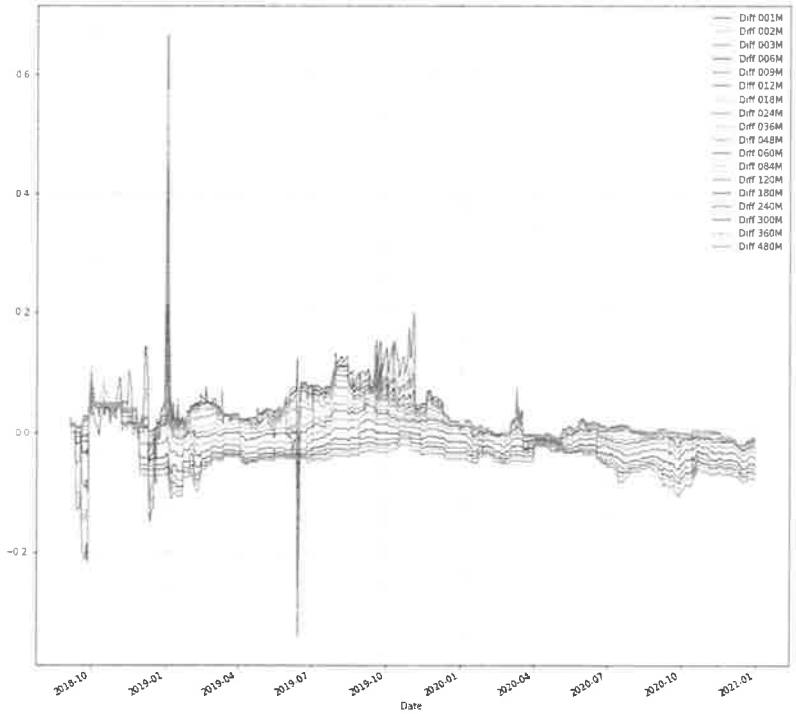
Training, Validation, Testing Period SOFR Summary Statistics: 60M – 480M

Tenor	60M	84M	120M	180M	240M	300M	360M	480M
Median								
Test	2.641	2.643	2.664	2.690	2.692	2.675	2.655	2.610
Train	0.413	0.512	0.733	0.972	1.087	1.134	1.161	1.117
Validation	2.095	2.134	2.219	2.331	2.378	2.391	2.395	2.376
Maximum								
Test	2.927	2.954	2.995	3.030	3.037	3.028	3.012	2.969
Train	1.967	2.000	2.082	2.185	2.232	2.244	2.246	2.230
Validation	2.336	2.373	2.458	2.557	2.591	2.596	2.594	2.567
Mean								
Test	2.574	2.596	2.645	2.696	2.707	2.697	2.682	2.643
Train	0.749	0.839	0.968	1.101	1.169	1.192	1.200	1.165
Validation	2.112	2.149	2.232	2.336	2.379	2.388	2.390	2.367
Minimum								
Test	2.150	2.185	2.258	2.330	2.347	2.345	2.339	2.302
Train	0.022	0.125	0.262	0.358	0.375	0.369	0.354	0.303
Validation	1.920	1.965	2.058	2.168	2.216	2.229	2.231	2.210
Standard Deviation								
Test	0.236	0.235	0.227	0.218	0.215	0.212	0.209	0.207
Train	0.613	0.567	0.528	0.505	0.497	0.493	0.490	0.497
Validation	0.105	0.103	0.103	0.102	0.098	0.097	0.096	0.093

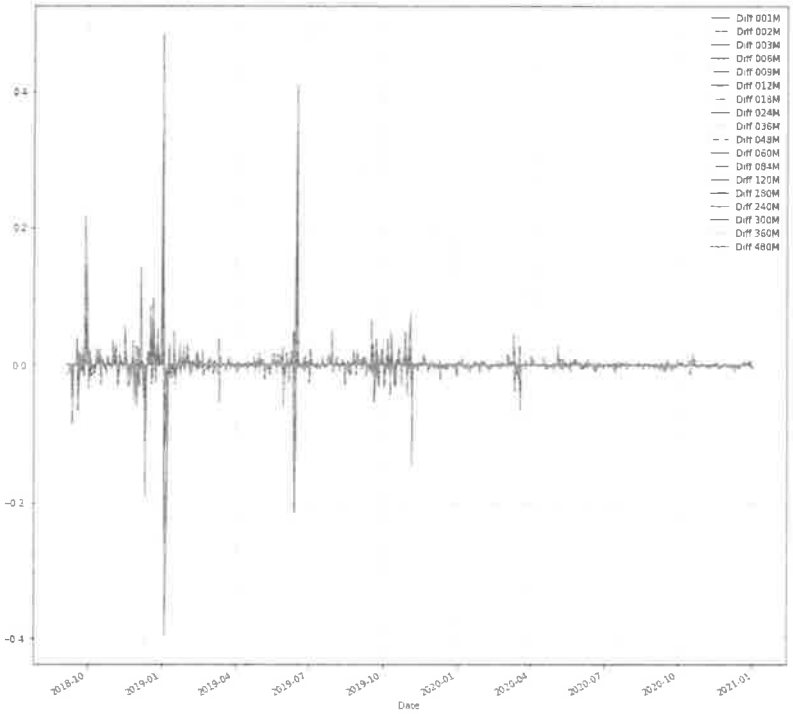
Summary statistics for SOFR 60M – 480M per sample period. Test sample is 2018-09-04 to 2019-02-25 (T=116, 20.10%), validation sample is 2019-02-26 to 2019-05-21 (T=58, 10.05%), train sample is 2019-05-22 – 2020-12-31 (T=403, 69.84%). All tenors non-stationary for all samples.

Time-Series Properties of SOFR-EFFR Spread

SOFR – EFFR Spreads, 9/4/18 – 12/31/20 Per Tenor



First Differenced SOFR-EFFR Spread Per tenor, 9/4/18 – 12/31/20



9/4/18+ SOFR-EFFR Spread ADF Results Per Tenor

Tenor	Test Statistic	P-Value	Lags	T	Reject H_0 at 5% Level	Conc.
001M	-3.841	0.003	14	561	True	Stationary
002M	-3.855	0.002	14	561	True	Stationary
003M	-3.130	0.024	14	561	True	Stationary
006M	-1.964	0.302	14	561	False	Non-Stationary
009M	-1.673	0.445	14	561	False	Non-Stationary
012M	-1.561	0.503	14	561	False	Non-Stationary
018M	-1.383	0.590	14	561	False	Non-Stationary
024M	-1.451	0.558	13	562	False	Non-Stationary
036M	-1.462	0.552	13	562	False	Non-Stationary
048M	-1.557	0.505	13	562	False	Non-Stationary
060M	-1.681	0.441	13	562	False	Non-Stationary
084M	-1.782	0.389	9	566	False	Non-Stationary
120M	-2.300	0.172	14	561	False	Non-Stationary
180M	-2.635	0.086	14	561	False	Non-Stationary
240M	-2.779	0.061	14	561	False	Non-Stationary
300M	-2.883	0.047	14	561	True	Stationary
360M	-2.958	0.039	14	561	True	Stationary
480M	-3.027	0.032	14	561	True	Stationary

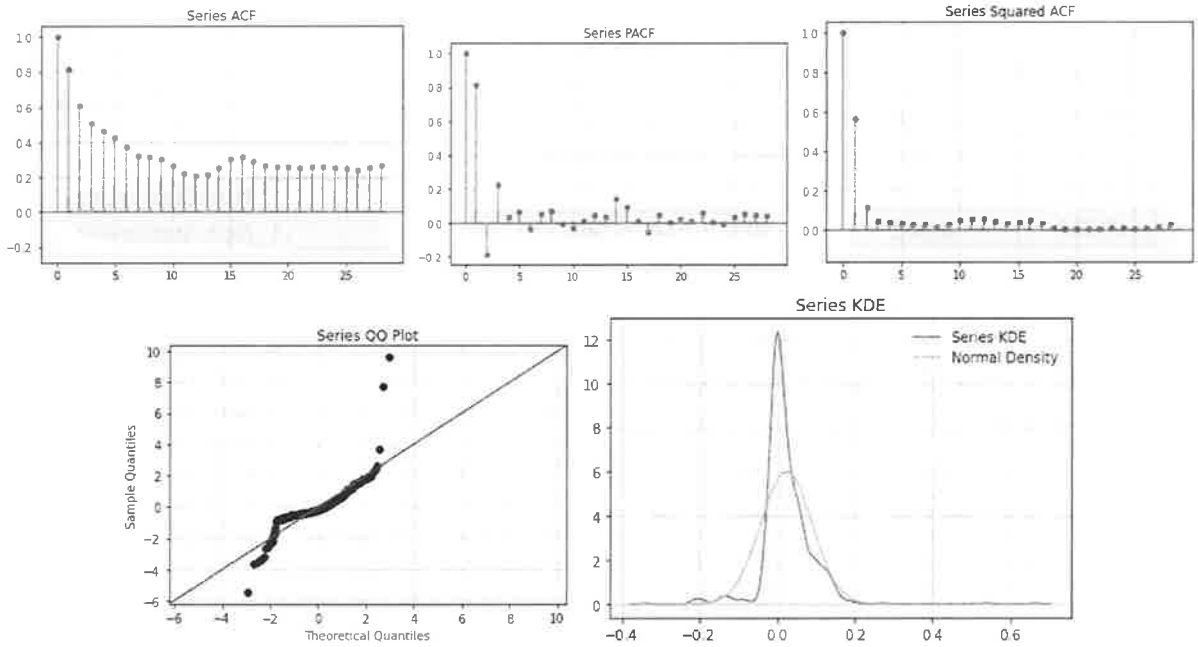
H_0: series contains unit root, meaning non-stationary in levels but FD stationary. CVs: 10%, -3.442, 5%, -2.867, 1%, -2.570

9/4/18+ SOFR-EFFR FD Spread ADF Results Per Tenor

Tenor	Test Statistic	P-Value	Lags	T	Reject H_0 at 5% Level	Conc.
001M	-11.363	0.000	13	561	True	Stationary
002M	-10.843	0.000	13	561	True	Stationary
003M	-10.679	0.000	13	561	True	Stationary
006M	-9.751	0.000	13	561	True	Stationary
009M	-8.925	0.000	13	561	True	Stationary
012M	-8.468	0.000	13	561	True	Stationary
018M	-8.131	0.000	13	561	True	Stationary
024M	-7.950	0.000	12	562	True	Stationary
036M	-7.771	0.000	12	562	True	Stationary
048M	-7.485	0.000	12	562	True	Stationary
060M	-7.160	0.000	12	562	True	Stationary
084M	-4.936	0.000	19	555	True	Stationary
120M	-5.038	0.000	19	555	True	Stationary
180M	-4.991	0.000	19	555	True	Stationary
240M	-4.972	0.000	19	555	True	Stationary
300M	-4.940	0.000	19	555	True	Stationary
360M	-4.912	0.000	19	555	True	Stationary
480M	-5.643	0.000	13	561	True	Stationary

H_0: series contains unit root, meaning non-stationary in levels but FD stationary. CVs: 10%, -3.442, 5%, -2.867, 1%, -2.570

1M Spread Diagnostics, 9/4/18 + Data



LB Tests on SOFR-EFFR Spread, Levels and first difference, 9/4/18+ data

Tenor	Level			FD		
	LB Stat.	P-Value	Conc.	LB Stat.	P-Value	Conc.
001M	975.261	0.000	Reject H0	73.687	0.000	Reject H0
002M	968.903	0.000	Reject H0	70.316	0.000	Reject H0
003M	1,071.982	0.000	Reject H0	64.273	0.000	Reject H0
006M	1,895.579	0.000	Reject H0	44.464	0.000	Reject H0
009M	2,212.029	0.000	Reject H0	37.438	0.000	Reject H0
012M	2,343.710	0.000	Reject H0	31.191	0.000	Reject H0
018M	2,478.969	0.000	Reject H0	22.278	0.000	Reject H0
024M	2,523.987	0.000	Reject H0	15.665	0.008	Reject H0
036M	2,571.402	0.000	Reject H0	9.142	0.104	Fail to Reject
048M	2,597.939	0.000	Reject H0	4.072	0.539	Fail to Reject
060M	2,630.654	0.000	Reject H0	2.969	0.705	Fail to Reject
084M	2,676.298	0.000	Reject H0	2.534	0.771	Fail to Reject
120M	2,682.998	0.000	Reject H0	1.772	0.880	Fail to Reject
180M	2,658.765	0.000	Reject H0	3.983	0.552	Fail to Reject
240M	2,646.814	0.000	Reject H0	9.346	0.096	Fail to Reject
300M	2,640.702	0.000	Reject H0	15.390	0.009	Reject H0
360M	2,637.908	0.000	Reject H0	21.782	0.001	Reject H0
480M	2,634.671	0.000	Reject H0	30.917	0.000	Reject H0

ABM Backfilled SOFR Spread (1/3/7 – 9/4/18) Stationarity Tests

9/4/18 SOFR - EFFR Spread ADF (Levels)

Tenor	Test Statistic	P-Value	Lags	T	Reject H_0 at 5%	Conc.
001M	-1.597	0.485	27	2902	False	Non-Stationary
002M	-1.726	0.418	0	2929	False	Non-Stationary
003M	-1.782	0.389	0	2929	False	Non-Stationary
006M	-2.683	0.077	19	2910	False	Non-Stationary
009M	-3.309	0.014	19	2910	True	Stationary
012M	-3.194	0.020	19	2910	True	Stationary
018M	-2.678	0.078	19	2910	False	Non-Stationary
024M	-2.062	0.260	19	2910	False	Non-Stationary
036M	-1.415	0.575	19	2910	False	Non-Stationary
048M	-0.796	0.820	26	2903	False	Non-Stationary
060M	-0.590	0.873	26	2903	False	Non-Stationary
084M	-1.136	0.701	26	2903	False	Non-Stationary
120M	-1.808	0.376	19	2910	False	Non-Stationary
180M	-2.375	0.149	19	2910	False	Non-Stationary
240M	-2.190	0.210	25	2904	False	Non-Stationary
300M	-2.031	0.273	25	2904	False	Non-Stationary
360M	-1.683	0.440	25	2904	False	Non-Stationary
480M	-1.613	0.477	0	2929	False	Non-Stationary

9/4/18 SOFR - EFFR Spread ADF (FD)

Tenor	Test Statistic	P-Value	Lags	T	Reject H_0 at 5%	Conc.
001M	-10.595	0.000	26	2902	True	Stationary
002M	-54.668	0.000	0	2928	True	Stationary
003M	-54.234	0.000	0	2928	True	Stationary
006M	-11.443	0.000	18	2910	True	Stationary
009M	-11.344	0.000	18	2910	True	Stationary
012M	-11.352	0.000	18	2910	True	Stationary
018M	-11.187	0.000	18	2910	True	Stationary
024M	-10.901	0.000	18	2910	True	Stationary
036M	-10.809	0.000	18	2910	True	Stationary
048M	-9.452	0.000	25	2903	True	Stationary
060M	-9.286	0.000	25	2903	True	Stationary
084M	-9.183	0.000	25	2903	True	Stationary
120M	-11.003	0.000	18	2910	True	Stationary
180M	-11.613	0.000	18	2910	True	Stationary
240M	-13.319	0.000	14	2914	True	Stationary
300M	-9.702	0.000	24	2904	True	Stationary
360M	-54.726	0.000	0	2928	True	Stationary
480M	-55.111	0.000	0	2928	True	Stationary

ABM Random Draw Variation

I outline the process for non-parametric bootstrapping of test error measure standard errors and present summary statistics per tenor. The arithmetic Brownian motion and white noise processes for backfilling dynamic spreads require repeatedly drawing from a multivariate normal distribution, where draws across time are uncorrelated but cross-tenor spreads at a single time are correlated. By summing these draws, the ABM process produces spreads and thus backfilled SOFR that is highly susceptible to variability in the drawn values. It is possible to fix the draws for the purposes of reproducibility by setting the random seed. I exploit this property by repeating the process to generate dynamic spreads 1,000 times, and for each varying the random seed to guarantee a different set of random draws. For each random seed, the process yields backfilled spreads and SOFR, and I compute error measures on the test set (R^2 and RMSE) and retain them. The result is 1,000 test R^2 and RMSE values per tenor (1 month to 600 months) for the spreads and 1,000 test R^2 and RMSE values per tenor for backfilled SOFR. The standard error of the average test error measure is the standard deviation of the empirical sampling distribution of the average test error measure and equals the standard deviation of the population of test error measures divided by the $1{,}000^{0.5}$. I report the average, standard deviation, empirical 95% confidence interval, and standard error of the average for SOFR and spread test R^2 and RMSE below.

ABM Test Spread R^2 Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
Sp001M	-0.099	0.102	-0.503	-0.337	-0.086	0.061	0.101	0.003
Sp002M	-0.090	0.102	-0.572	-0.338	-0.071	0.053	0.087	0.003
Sp003M	-0.215	0.185	-1.123	-0.655	-0.187	0.047	0.095	0.006
Sp006M	-0.647	0.428	-3.167	-1.656	-0.579	-0.019	0.142	0.014
Sp009M	-1.010	0.329	-2.668	-1.709	-0.986	-0.445	-0.161	0.010
Sp012M	-1.296	0.249	-2.415	-1.805	-1.287	-0.840	-0.565	0.008
Sp018M	-0.811	0.109	-1.255	-1.025	-0.807	-0.610	-0.455	0.003
Sp024M	-0.263	0.039	-0.414	-0.341	-0.262	-0.197	-0.132	0.001
Sp036M	-0.123	0.025	-0.226	-0.173	-0.123	-0.076	-0.044	0.001
Sp048M	-0.674	0.027	-0.781	-0.726	-0.674	-0.622	-0.581	0.001
Sp060M	-1.158	0.023	-1.252	-1.201	-1.158	-1.114	-1.088	0.001
Sp084M	-1.402	0.013	-1.455	-1.426	-1.402	-1.376	-1.357	0.000
Sp120M	-1.035	0.006	-1.054	-1.047	-1.035	-1.022	-1.014	0.000
Sp180M	-0.619	0.004	-0.629	-0.626	-0.619	-0.612	-0.607	0.000
Sp240M	-0.491	0.004	-0.504	-0.500	-0.492	-0.482	-0.475	0.000
Sp300M	-0.425	0.004	-0.434	-0.432	-0.425	-0.418	-0.412	0.000
Sp360M	-0.385	0.002	-0.392	-0.390	-0.386	-0.381	-0.378	0.000
Sp480M	-0.340	0.003	-0.348	-0.345	-0.340	-0.335	-0.332	0.000
Sp600M	-0.316	0.003	-0.327	-0.323	-0.316	-0.310	-0.306	0.000

Summary statistics for distribution of backfilled SOFR-EFFR spread R^2s per tenor assuming ABM process and generated by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

ABM Test Spread RMSE Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
Sp001M	0.115	0.005	0.105	0.107	0.115	0.127	0.135	0.000
Sp002M	0.095	0.004	0.087	0.089	0.095	0.106	0.115	0.000
Sp003M	0.076	0.006	0.066	0.067	0.075	0.089	0.100	0.000
Sp006M	0.044	0.006	0.032	0.035	0.044	0.057	0.071	0.000
Sp009M	0.037	0.003	0.028	0.032	0.037	0.043	0.051	0.000
Sp012M	0.034	0.002	0.028	0.031	0.034	0.038	0.042	0.000
Sp018M	0.025	0.001	0.022	0.024	0.025	0.026	0.028	0.000
Sp024M	0.018	0.000	0.017	0.018	0.018	0.019	0.020	0.000
Sp036M	0.017	0.000	0.017	0.017	0.017	0.018	0.018	0.000
Sp048M	0.023	0.000	0.022	0.022	0.023	0.023	0.024	0.000
Sp060M	0.028	0.000	0.028	0.028	0.028	0.029	0.029	0.000
Sp084M	0.037	0.000	0.036	0.036	0.037	0.037	0.037	0.000
Sp120M	0.045	0.000	0.045	0.045	0.045	0.045	0.045	0.000
Sp180M	0.054	0.000	0.054	0.054	0.054	0.054	0.054	0.000
Sp240M	0.059	0.000	0.059	0.059	0.059	0.059	0.059	0.000
Sp300M	0.063	0.000	0.062	0.062	0.063	0.063	0.063	0.000
Sp360M	0.065	0.000	0.065	0.065	0.065	0.065	0.065	0.000
Sp480M	0.068	0.000	0.068	0.068	0.068	0.068	0.068	0.000
Sp600M	0.070	0.000	0.069	0.069	0.070	0.070	0.070	0.000

Summary statistics for distribution of backfilled SOFR-EFFR spread R^2s per tenor assuming ABM process and generated by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

ABM Test SOFR R^2 Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
001M	0.583	0.039	0.429	0.492	0.588	0.644	0.659	0.001
002M	0.617	0.036	0.448	0.530	0.624	0.667	0.679	0.001
003M	0.659	0.052	0.405	0.536	0.667	0.733	0.746	0.002
006M	0.647	0.092	0.107	0.431	0.662	0.782	0.816	0.003
009M	0.515	0.079	0.115	0.346	0.521	0.651	0.720	0.003
012M	0.723	0.030	0.588	0.662	0.724	0.778	0.811	0.001
018M	0.955	0.003	0.944	0.950	0.956	0.960	0.964	0.000
024M	0.987	0.000	0.986	0.987	0.987	0.988	0.989	0.000
036M	0.994	0.000	0.993	0.993	0.994	0.994	0.994	0.000
048M	0.990	0.000	0.990	0.990	0.990	0.991	0.991	0.000
060M	0.985	0.000	0.985	0.985	0.985	0.986	0.986	0.000
084M	0.976	0.000	0.975	0.975	0.976	0.976	0.976	0.000
120M	0.961	0.000	0.960	0.960	0.961	0.961	0.961	0.000
180M	0.938	0.000	0.938	0.938	0.938	0.939	0.939	0.000
240M	0.924	0.000	0.923	0.923	0.924	0.924	0.925	0.000
300M	0.913	0.000	0.912	0.912	0.913	0.913	0.913	0.000
360M	0.903	0.000	0.903	0.903	0.903	0.904	0.904	0.000
480M	0.891	0.000	0.891	0.891	0.891	0.892	0.892	0.000
600M	0.884	0.000	0.883	0.883	0.884	0.884	0.885	0.000

Summary statistics for distribution of backfilled SOFR R^2s per tenor assuming ABM process and generated by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

ABM Test SOFR RMSE Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
001M	0.115	0.005	0.105	0.107	0.115	0.127	0.135	0.000
002M	0.095	0.004	0.087	0.089	0.095	0.106	0.115	0.000
003M	0.076	0.006	0.066	0.067	0.075	0.089	0.100	0.000
006M	0.044	0.006	0.032	0.035	0.044	0.057	0.071	0.000
009M	0.037	0.003	0.028	0.032	0.037	0.043	0.051	0.000
012M	0.034	0.002	0.028	0.031	0.034	0.038	0.042	0.000
018M	0.025	0.001	0.022	0.024	0.025	0.026	0.028	0.000
024M	0.018	0.000	0.017	0.018	0.018	0.019	0.020	0.000
036M	0.017	0.000	0.017	0.017	0.017	0.018	0.018	0.000
048M	0.023	0.000	0.022	0.022	0.023	0.023	0.024	0.000
060M	0.028	0.000	0.028	0.028	0.028	0.029	0.029	0.000
084M	0.037	0.000	0.036	0.036	0.037	0.037	0.037	0.000
120M	0.045	0.000	0.045	0.045	0.045	0.045	0.045	0.000
180M	0.054	0.000	0.054	0.054	0.054	0.054	0.054	0.000
240M	0.059	0.000	0.059	0.059	0.059	0.059	0.059	0.000
300M	0.063	0.000	0.062	0.062	0.063	0.063	0.063	0.000
360M	0.065	0.000	0.065	0.065	0.065	0.065	0.065	0.000
480M	0.068	0.000	0.068	0.068	0.068	0.068	0.068	0.000
600M	0.070	0.000	0.069	0.069	0.070	0.070	0.070	0.000

Summary statistics for distribution of backfilled SOFR R^2s per tenor assuming ABM process and generated by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

White Noise Test Spread R^2 Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
Sp001M	-0.216	0.089	-0.543	-0.397	-0.216	-0.042	0.062	0.003
Sp002M	-0.219	0.091	-0.530	-0.401	-0.219	-0.045	0.073	0.003
Sp003M	-0.328	0.114	-0.729	-0.555	-0.330	-0.115	0.064	0.004
Sp006M	-0.994	0.225	-1.726	-1.458	-0.987	-0.571	-0.384	0.007
Sp009M	-1.483	0.298	-2.511	-2.079	-1.472	-0.903	-0.654	0.009
Sp012M	-1.679	0.326	-2.773	-2.373	-1.671	-1.033	-0.764	0.010
Sp018M	-1.899	0.358	-3.039	-2.633	-1.889	-1.224	-0.836	0.011
Sp024M	-1.973	0.365	-3.111	-2.721	-1.962	-1.305	-0.773	0.012
Sp036M	-1.613	0.307	-2.622	-2.220	-1.598	-1.060	-0.572	0.010
Sp048M	-1.311	0.247	-2.236	-1.808	-1.299	-0.849	-0.527	0.008
Sp060M	-1.180	0.211	-2.029	-1.629	-1.173	-0.778	-0.559	0.007
Sp084M	-0.981	0.166	-1.662	-1.327	-0.976	-0.669	-0.551	0.005
Sp120M	-0.619	0.116	-1.055	-0.860	-0.615	-0.399	-0.272	0.004
Sp180M	-0.372	0.080	-0.616	-0.528	-0.370	-0.222	-0.123	0.003
Sp240M	-0.318	0.069	-0.537	-0.456	-0.315	-0.189	-0.111	0.002
Sp300M	-0.306	0.065	-0.510	-0.437	-0.304	-0.187	-0.112	0.002
Sp360M	-0.308	0.064	-0.500	-0.435	-0.308	-0.190	-0.109	0.002
Sp480M	-0.313	0.065	-0.510	-0.443	-0.314	-0.194	-0.104	0.002
Sp600M	-0.320	0.067	-0.529	-0.452	-0.321	-0.200	-0.105	0.002

Summary statistics for distribution of backfilled SOFR-EFFR spread R^2s per tenor assuming white noise process generated by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

White Noise Test Spread RMSE Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
Sp001M	0.121	0.004	0.107	0.113	0.122	0.130	0.137	0.000
Sp002M	0.101	0.004	0.088	0.094	0.101	0.108	0.113	0.000
Sp003M	0.079	0.003	0.067	0.073	0.080	0.086	0.091	0.000
Sp006M	0.049	0.003	0.041	0.044	0.049	0.055	0.058	0.000
Sp009M	0.042	0.003	0.034	0.036	0.042	0.046	0.049	0.000
Sp012M	0.037	0.002	0.030	0.032	0.037	0.041	0.044	0.000
Sp018M	0.032	0.002	0.025	0.028	0.032	0.035	0.037	0.000
Sp024M	0.028	0.002	0.022	0.025	0.028	0.032	0.033	0.000
Sp036M	0.026	0.002	0.020	0.023	0.026	0.029	0.031	0.000
Sp048M	0.027	0.001	0.022	0.024	0.027	0.030	0.032	0.000
Sp060M	0.029	0.001	0.024	0.026	0.029	0.031	0.034	0.000
Sp084M	0.033	0.001	0.029	0.031	0.033	0.036	0.039	0.000
Sp120M	0.040	0.001	0.035	0.037	0.040	0.043	0.045	0.000
Sp180M	0.049	0.001	0.045	0.047	0.049	0.052	0.054	0.000
Sp240M	0.056	0.001	0.051	0.053	0.056	0.058	0.060	0.000
Sp300M	0.060	0.001	0.055	0.057	0.060	0.063	0.064	0.000
Sp360M	0.063	0.002	0.058	0.060	0.063	0.066	0.067	0.000
Sp480M	0.067	0.002	0.062	0.064	0.067	0.070	0.072	0.000
Sp600M	0.070	0.002	0.064	0.066	0.070	0.073	0.075	0.000

Summary statistics for distribution of backfilled SOFR-EFFR spread RMSE per tenor assuming white noise process by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

White Noise Test SOFR R^2 Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
001M	0.538	0.034	0.414	0.470	0.538	0.605	0.644	0.001
002M	0.572	0.032	0.462	0.508	0.572	0.633	0.674	0.001
003M	0.627	0.032	0.515	0.564	0.627	0.687	0.737	0.001
006M	0.573	0.048	0.416	0.473	0.574	0.663	0.703	0.002
009M	0.401	0.072	0.153	0.257	0.403	0.541	0.601	0.002
012M	0.677	0.039	0.545	0.593	0.678	0.755	0.787	0.001
018M	0.929	0.009	0.901	0.911	0.929	0.945	0.955	0.000
024M	0.970	0.004	0.959	0.963	0.971	0.977	0.982	0.000
036M	0.985	0.002	0.979	0.982	0.985	0.988	0.991	0.000
048M	0.987	0.001	0.981	0.984	0.987	0.989	0.991	0.000
060M	0.985	0.001	0.979	0.982	0.985	0.988	0.989	0.000
084M	0.980	0.002	0.973	0.976	0.980	0.983	0.984	0.000
120M	0.969	0.002	0.960	0.964	0.969	0.973	0.975	0.000
180M	0.948	0.003	0.939	0.942	0.948	0.954	0.957	0.000
240M	0.933	0.004	0.921	0.926	0.933	0.939	0.943	0.000
300M	0.920	0.004	0.907	0.912	0.920	0.927	0.932	0.000
360M	0.909	0.004	0.895	0.900	0.909	0.917	0.923	0.000
480M	0.894	0.005	0.878	0.883	0.893	0.903	0.910	0.000
600M	0.883	0.006	0.865	0.872	0.883	0.894	0.902	0.000

Summary statistics for distribution of backfilled SOFR R^2s per tenor assuming white noise process and generated by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

White Noise Test SOFR RMSE Distribution

Tenor	Mean	Std.	Min.	P2.5	P50	P97.5	Max	SE(Mean)
001M	0.121	0.004	0.107	0.113	0.122	0.130	0.137	0.000
002M	0.101	0.004	0.088	0.094	0.101	0.108	0.113	0.000
003M	0.079	0.003	0.067	0.073	0.080	0.086	0.091	0.000
006M	0.049	0.003	0.041	0.044	0.049	0.055	0.058	0.000
009M	0.042	0.003	0.034	0.036	0.042	0.046	0.049	0.000
012M	0.037	0.002	0.030	0.032	0.037	0.041	0.044	0.000
018M	0.032	0.002	0.025	0.028	0.032	0.035	0.037	0.000
024M	0.028	0.002	0.022	0.025	0.028	0.032	0.033	0.000
036M	0.026	0.002	0.020	0.023	0.026	0.029	0.031	0.000
048M	0.027	0.001	0.022	0.024	0.027	0.030	0.032	0.000
060M	0.029	0.001	0.024	0.026	0.029	0.031	0.034	0.000
084M	0.033	0.001	0.029	0.031	0.033	0.036	0.039	0.000
120M	0.040	0.001	0.035	0.037	0.040	0.043	0.045	0.000
180M	0.049	0.001	0.045	0.047	0.049	0.052	0.054	0.000
240M	0.056	0.001	0.051	0.053	0.056	0.058	0.060	0.000
300M	0.060	0.001	0.055	0.057	0.060	0.063	0.064	0.000
360M	0.063	0.002	0.058	0.060	0.063	0.066	0.067	0.000
480M	0.067	0.002	0.062	0.064	0.067	0.070	0.072	0.000
600M	0.070	0.002	0.064	0.066	0.070	0.073	0.075	0.000

Summary statistics for distribution of backfilled SOFR RMSE per tenor assuming white noise process and generated by varying the random seed. N=1,000 iterations. P2.5 and P97.5 comprise 95% confidence interval. Standard error of mean is standard deviation divided by $1000^{0.5}$.

PCR w/Z-Scoring Yields Results

Standardized PCR Results

PCR Model Error, 1M – 36M

Tenor	1M	2M	3M	6M	9M	12M	18M	24M	36M
M*	2	2	2	2	1	5	4	4	4
Train									
R^2	0.983	0.989	0.992	0.997	0.979	1.000	1.000	1.000	0.999
MAPE	0.134	0.097	0.079	0.053	0.134	0.023	0.027	0.036	0.025
MSE	0.017	0.011	0.008	0.003	0.021	0.000	0.000	0.000	0.001
RMSE	0.130	0.105	0.088	0.056	0.145	0.021	0.019	0.022	0.025
Validation									
R^2	0.698	0.803	0.827	0.923	0.945	0.957	0.987	0.992	0.995
MAPE	2.227	4.514	1.987	0.897	1.523	0.925	0.493	0.466	0.241
MSE	0.297	0.194	0.170	0.075	0.054	0.042	0.013	0.008	0.005
RMSE	0.545	0.441	0.412	0.274	0.232	0.205	0.114	0.087	0.069
Test									
R^2	0.054	0.048	0.065	0.273	0.326	0.739	0.791	0.867	0.950
MAPE	5.313	2.364	5.746	1.353	3.384	0.890	1.042	0.914	0.965
MSE	0.938	0.944	0.927	0.721	0.668	0.259	0.207	0.132	0.049
RMSE	0.969	0.972	0.963	0.849	0.817	0.509	0.455	0.363	0.222

Training, validation, and test period error measures for 1M – 36M PCR models. M* per tenor is the optimal number of principal components per validation set $R^2_{Adj.}$. I report R^2 (not adjusted), MAPE, MSE, and RMSE. MAPE over 1 means 100%+. Training period: 2019-05-22 – 2020-12-31 (T=403, 69.84%), Validation period: 2019-02-26 – 2019-05-21 (T=58, 10.05%), Test period: 2018-09-04 – 2019-02-25 (T=116, 20.10%).

PCR Model Error, 48M – 600M

Tenor	48M	60M	84M	120M	180m	240m	300M	360M	480M	600M
M*	4	4	4	5	2	4	3	4	4	2
Train										
R^2	4	1.000	1.000	1.000	0.999	1.000	1.000	0.999	0.999	0.997
MAPE	0.022	0.021	0.020	0.024	0.037	0.028	0.039	0.057	0.103	0.088
MSE	0.001	0.000	0.000	0.000	0.001	0.000	0.000	0.001	0.001	0.003
RMSE	0.023	0.021	0.019	0.020	0.028	0.018	0.022	0.025	0.032	0.054
Validation										
R^2	0.998	0.998	0.996	0.993	0.997	0.998	0.998	0.998	0.996	0.991
MAPE	0.110	0.096	0.178	0.239	0.162	0.237	0.119	0.166	0.172	0.235
MSE	0.002	0.002	0.004	0.006	0.003	0.002	0.002	0.002	0.004	0.009
RMSE	0.043	0.044	0.063	0.080	0.053	0.044	0.044	0.049	0.061	0.093
Test										
R^2	0.988	0.997	0.986	0.989	0.991	0.992	0.989	0.983	0.973	0.968
MAPE	0.231	0.077	1.126	0.188	0.215	0.165	0.211	0.341	0.727	0.671
MSE	0.011	0.003	0.014	0.011	0.009	0.008	0.011	0.017	0.027	0.031
RMSE	0.107	0.057	0.119	0.105	0.095	0.090	0.103	0.130	0.164	0.177

Training, validation, and test period error measures for 48M – 600M PCR models. M* per tenor is the optimal number of principal components per validation set $R^2_{Adj.}$. I report R^2 (not adjusted), MAPE, MSE, and RMSE. MAPE over 1 means 100%+. Training period: 2019-05-22 – 2020-12-31 (T=403, 69.84%), Validation period: 2019-02-26 – 2019-05-21 (T=58, 10.05%), Test period: 2018-09-04 – 2019-02-25 (T=116, 20.10%).

Full Sample Regression Test Results

Regression Test: Backfilled 1M SOFR on Survey and GCF Rates, 2007 - 2016

	1M No Spread SOFR	1M Avg. Spread SOFR	1M MVN Spread SOFR	1M PCR SOFR	1M ABM Spread SOFR
Constant	-0.0254 (0.0157)	-0.0016 (0.0157)	-0.0023 (0.0159)	-0.1379*** (0.0181)	1.2054*** (0.0436)
Survey Rate	-0.1811 (0.3713)	-0.1811 (0.3713)	-0.1844 (0.3732)	-1.0653*** (0.3723)	3.6385*** (0.7845)
Treasury GCF Repo Weighted Average Rate	1.1741*** (0.3737)	1.1741*** (0.3737)	1.1781*** (0.3757)	2.0848*** (0.3747)	-2.5775*** (0.7887)
R^2	0.9810	0.9810	0.9788	0.9771	0.9031

Coefficient estimates for $\widehat{SOFR}_t = \beta_0 + \beta_1 SurveyRate_t + \beta_2 GCF_t + \varepsilon_t$. T=2,441, 01/03/07 – 10/31/16. Standard errors in parenthesis, stars indicate significance: ***: 1%, **, 5%, *:10%. Newey-West standard errors, $T^{0.25}$ lags.